

Mirroror

Manual

Version 1.0

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0. About this instrument

Nothing needs to be on screen for this chapter. Every chapter after it assumes the app is open.

What it is

Mirroror is a video synthesizer. A picture goes in, gets folded, twisted, and pushed out of shape, and comes out the other side. The picture comes from a camera, from a video file, or from a pattern the instrument generates itself.

All of it happens live. There is no render step and no offline pass. Every control is playable while the image is running: move a slider and the picture changes under your hand at sixty frames a second.



One frame out of the instrument.

00-01 One frame out of the instrument.

Geometry first

Every module moves the picture. It folds it across a line, twists it around a point, or pushes it off its own coordinates. Nothing paints and nothing draws on top.

There is no color control anywhere in the instrument. No hue, no saturation, no channel splitting. What color arrives is the color that leaves, put somewhere else. A fold sets one region's color against another's. A twist smears it around a center. Where two folds interfere, the image gets cut away and the background shows through.

Point a camera at something red and the only red in the output is that red, wherever the geometry has carried it.



The source, then the same source through the chain. No color control was touched, because there is not one.

00-02 The source, then the same source through the chain. No color control was touched, because there is not one.

What it runs on

macOS, drawing on the Mac's GPU. The frame rate is locked to 60.

The window opens with the picture on the left and a fixed control column on the right, and it will not go below 830 × 550 points. The picture area holds a minimum of 500 points on each side.

The camera is whatever camera macOS hands over, and the first time you select it macOS asks for permission. Video files are read from disk and loop. Recordings are written from the image the instrument makes rather than from the screen, at sizes up to 3840 × 2160, which is chapter 13.

There are no patches in the box

Nothing ships with a sound-alike patch. The buttons under Presets in **Geometry Order Routing** set chain order and nothing else: which module runs in which slot. Every amplitude, frequency, angle, and key is dialed by hand.

This manual is written to stand in for the patches that are not there. Every chapter ends with a block called **Starting points**: a patch given control by control, with one thing to move once it is running.

How to read this

Control names appear in bold, spelled the way the app spells them: **VCO Amplitude**, **Luma → Pivot**, **Dry / Effected**. Section names are bold and exact too: **Geometry Order Routing**, **Global LFO**.

Numbers are the numbers on screen. Every slider row shows its value to two decimals beside a reset arrow, so any figure in this manual can be dialed by eye. Ranges are written as 0 to 1. Ranges that cross zero are written as -1 to 1, and the manual says what the negative half does.

Each chapter opens with an italic line naming the patch it assumes and closes with **Starting points**. Chapters can be read in any order: where one needs something from another, it says so in a line and links.

The chapters group like this:

- **1 and 2** get a picture moving and then map the whole signal path.
- **3** covers what goes in, and the frame edge.
- **4 and 5** cover the chain the geometry modules sit in, and the oscillator block those modules share.
- **6, 7, and 8** are the geometry modules themselves: Mirrors, Rotation, Displacement.
- **9 and 10** cover what the chain feeds: interference, holes, and the three keyers.
- **11** is the Global LFO.
- **12 and 13** cover output shaping, recording, and capture.
- **14** covers MIDI.

- **15 and 16** are patches to build and symptoms to look up.
 - **17** is the glossary, **18** the index, and **Appendix A** lists every control with its range and default on one page.
-

Where to start

Chapter 1 gets an image moving and does not stop to explain itself. Chapter 2 is the map, and every later chapter points back at it. If you have patched a modular synth, chapter 5 is where the vocabulary gets pinned to what is on screen here.

1. First look

Before you start: launch the app. Nothing else is needed, and nothing here has to be undone afterward.

This chapter gets a picture moving and names the controls it touches. It does not explain what any of them did. That is chapter 2.

What is on screen

The window is in two parts: the picture on the left, a fixed control column on the right.

The window at launch. Picture on the left, controls on the right.

01-01 The window at launch. Picture on the left, controls on the right.

The column runs **Source, Canvas, Geometry Order Routing, Rotation 1, Rotation 2, Mirror 1, Mirror 2, Displacement, Keying, Global LFO, Mix, Vignette, Recording**. Every section is open at launch. Click a section's header to collapse it, which changes nothing about the picture.

That column order is a layout, not a signal order. Where the geometry modules run is set in **Geometry Order Routing**, and it can be changed while the image is running.

At the top of the column sit the instrument's name and **Reset All**.

A control row

Every slider is built the same way. The label sits on the left. On the right are a reset arrow, which returns that one control to its default, and the current value to two decimals. The track runs underneath.

One control row: label, reset arrow, value, track.

01-02 One control row: label, reset arrow, value, track.

The number moves as you drag, so every value printed in this manual can be dialed by eye.

Something is already running

At launch the **Source** is **Gradient Synth**, a color field the instrument generates for itself. It drifts on its own with nothing touched.

Below the five slot pickers in **Geometry Order Routing** a line reads back what the chain is doing. At launch it says:



Source → M2 → M1 → Rot 2 → Rot 1 → Output

Under it, a second line reads "Not in chain: Displacement".

Both mirrors are folding, at right angles to each other. **Rotation 1** has its **VCO Amplitude** at 0.5, which is the twist you can see turning the field around the middle of the frame. **Rotation 2** is in the chain with its **VCO Amplitude** at 0, so it passes the image through.

Cut it down to one module

In **Geometry Order Routing**, find the eight Preset buttons at the bottom of the section and press **M1 only**.



Geometry Order Routing after pressing M1 only.

01-03 Geometry Order Routing after pressing M1 only.

Slot 1 becomes Mirror 1 and the other four go Empty. The summary line now reads Source → M1 → Output. An Empty slot passes the image through untouched, so what is left is one mirror running on the source.

The twist is gone and the frame is symmetric about a vertical line down the middle. On the **Gradient Synth** field the fold reads as symmetry rather than as a hard edge: the color runs one way on the right, and the mirror of that on the left.

Turn one control

In **Mirror 1**, drag **VCO Amplitude** up to about 0.35.



VCO Amplitude at 0, then at 0.35. The straight fold becomes a wave.

01-04 VCO Amplitude at 0, then at 0.35. The straight fold becomes a wave.

The line down the middle stops being straight. It becomes a wave, and the wave is already traveling, because **Phase Speed** ships at 0.45 and that control sets a speed.

Three more moves, in the same module:

- **VCO Frequency** down from 5 to 2. Fewer bends, each one larger.
 - **VCO Frequency** up to 10. The bends pack in until they read as texture.
 - **Mirror Angle** slowly up from 0. The line rotates around the center of the frame and the wave rides around with it. Full travel is 6.28, which brings it back where it started.
-

Put your own picture in

Source offers three entries: **Gradient Synth**, **Live Camera**, and **Video File**.

Press **Live Camera**. The first time, macOS asks for camera access, and a status line under the picker reads "Requesting access..." until you answer. Once the camera is running the line reads Live, then the device name, then the resolution it is delivering. If the line stops anywhere else, it says what went wrong, and chapter 16 lists each one.



The same one-mirror patch, running on the camera.

01–05 The same one-mirror patch, running on the camera.

Video File puts a **Load Video...** button under the picker. Pick a file and it starts playing and loops.

The mirror settings do not change when the source does. The same fold is now running on whatever you fed it.

Getting back

Reset All, at the top of the column, returns every parameter to its default. The menu item is Reset All Parameters..., with the shortcut ⌘⇧R.

All three routes ask first. Confirming clears the current patch and leaves the source selection and any recording in progress alone.

Starting points

Two folds. From **M1 only**, set slot 2 to Mirror 2. Mirror 2 arrives folding at right angles to Mirror 1, so the frame goes to quadrants. Bring Mirror 2's **VCO Amplitude** up to about 0.35 to match Mirror 1 and the two waves cross each other along the way.

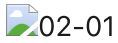
Hands off. Back to one mirror, **VCO Amplitude** at 0.35. Turn **Spin** on in **Mirror 1** and set **Spin Speed (Bipolar)** to about 0.2. The fold line turns on its own while the wave keeps traveling along it. Leave it running and read chapter 2 with it on screen.

2. The signal path

Nothing needs setting up for this chapter. Leave whatever chapter 1 left on screen and read it running.

This is the order the instrument works in. Every other chapter in this manual is one stage of it, so this is the page to come back to when a control does something you did not expect.

The whole path



Source, then the chain, then the hole cutter, then the keyers, then the mix, then the vignette, then out.

That order is fixed. The chain in the middle is the part you rearrange, and chapter 4 is about rearranging it. Everything on either side of it stays where it is.

Source

One picture goes in. **Gradient Synth**, **Live Camera**, or **Video File**, picked in **Source**.

Canvas shapes that picture before anything else touches it. **Zoom** sets how much of it the frame holds. **Border Behavior** decides what sits outside the picture once the geometry starts pulling coordinates past the edge. **Edge Pre-Crop (Raw Input)** trims the outermost pixels off the input first. Those three are properties of the source, not of any module, which is why they are in their own section and why they apply no matter what the chain does afterwards.

Chapter 3 covers all of it.

The chain

Five slots, in order. Each slot holds one geometry module or Empty.

The sidebar prints the direction under the slot pickers:

Signal order: slot 1 acts on the source, slot 5 is seen last

Slot 1 is the module closest to the source. Slot 5 is the module closest to your eye. A mirror in slot 1 folds the raw picture and everything after it smears that fold around. The same mirror in slot 5 folds the finished composite instead. Same module, same settings, different picture.

Under the pickers is a readback of what you have built. At launch it says:

Source → M2 → M1 → Rot 2 → Rot 1 → Output

with a second line beneath it:

Not in chain: Displacement

Empty slots do not appear in the readback. A module you have left out is named on the second line so you know it is sitting there with its controls live and nothing to act on. Empty every slot and the readback reads Source passes through unwarpd .

The same two modules, swapped between slots 1 and 2.

02-02 The same two modules, swapped between slots 1 and 2.

The five modules are Mirror 1 and Mirror 2 (chapter 6), Rotation 1 and Rotation 2 (chapter 7), and Displacement (chapter 8). A module can be listed in more than one slot, which runs it twice.

Everything after the chain reads what the chain made

The four stages after the chain do not see the source. They see the picture as the chain left it, so changing a slot changes what all four of them are working on.

The hole cutter

The two mirrors leave a wave interference pattern behind them. The hole cutter thresholds that pattern and cuts the image away where it crosses, so the background shows through. **Collision Threshold** sets where, **Wave Folding** puts more crossings in, and **Rotation → Holes** feeds the rotation wave into the same signal.

The pattern comes from the mirrors, so a chain with no mirror in it has nothing here to threshold. Chapter 9 says what happens then.

The keyers

Three keys, one per module family. **Keyer 1 — Mirrors**, **Keyer 2 — Rotation**, and **Keyer 3 — Displacement**. Each one decides which parts of the picture get cut away by brightness, and each one has a Feed picker that chooses whether it reads its own family or the finished warp.

The whole section is off at launch. Turn **Enabled** on at the top of **Keying** before any of it does anything. Chapter 10 is the one to read carefully, because the keyers are where the instrument stops being symmetrical.

The mix

Dry / Effected in **Mix** blends the finished picture back against the source, 0 to 1, default 1.

The dry side is the source with **Zoom**, **Border Behavior** and **Edge Pre-Crop (Raw Input)** applied and no geometry at all. Pulling the slider down brings the unwarped picture up underneath everything the chain, the holes, and the keys have done to it.

The vignette

A shape matte over the mixed result. **Off**, **Circle**, or **Rect**, with center, size, stretch, and softness. Outside the shape goes black.

It runs last, on the already-mixed image, so it crops the dry picture and the effected picture together. Nothing downstream of it exists.

What comes out

The window fits the picture into the space left of the sidebar. If the picture's aspect and the window's do not match, you get black bars on screen.

The recorder takes what the instrument makes, not what is on the screen. A take is written at whatever **Output Resolution** says, and on the fixed presets that size has nothing to do with how big the window is. The bars are not in the file. Neither is the sidebar, the cursor, or anything that passes in front of the window while you record. Screenshots come from the same place.

Two things follow from that. A recording is not a screen capture and cannot be compared to one. And the vignette is inside the picture rather than over it, so **Matte to Alpha** writes a real alpha channel into a take even though the window on screen stays opaque. Chapter 13 has the rest.

Starting points

Move one module and watch four stages change. Mirror 1 in slot 1, Rotation 1 in slot 2, everything else Empty. Set the mirror's **VCO Amplitude** to about 0.35 and Rotation 1's to about 0.5. Now swap the two slots. The geometry changes, and so does the pattern the hole cutter reads, because the mirror is now folding a twisted picture instead of feeding a twist.

Find the dry image. Same patch. Pull **Dry / Effected** down to about 0.4 and leave it. Every change you make to the chain from here is happening on top of a picture you can still recognise, which makes the geometry easier to read while you learn it. Put it back to 1 when you are done.

3. Source and Canvas

*Before you start: set every slot in **Geometry Order Routing** to Empty. The readback should say Source passes through unwarped . This chapter is about the picture before any module touches it.*

The source

The source is the one picture that goes in. **Source** offers three, as a segmented picker at the top of the column.

Gradient Synth

The instrument's own picture, and the one it opens with. Three color fields drifting at different speeds across each other. It has no controls.

The Gradient Synth is defined everywhere in the frame, so it never runs out of picture. The geometry can push coordinates as far outside the frame as it likes and still find color there, which means **Border Behavior** and **Edge Pre-Crop (Raw Input)** below have nothing to act on. Both do nothing while this source is selected.

A hard fold on it reads as symmetry rather than as a cut, because there is no recognisable subject to notice has been replaced.

Live Camera

Selecting it starts the camera. The first time, macOS asks for permission.

A status line appears under the picker and reports the sequence:

Requesting access...

Starting...

Live – FaceTime HD Camera – 1280x720

An external camera is taken over the built in one when both are connected. The app asks the device for 1280 x 720 and falls back to the device's own default where that is not available, which the line names instead of the size.

If the line stops anywhere else it turns red and says what went wrong. Chapter 16 has each one.

The Source section with the camera running.

03-01 The Source section with the camera running.

Video File

Puts a **Load Video...** button under the picker. Pick a file and it starts playing and loops back to the beginning without a gap. The button then reads the file's name.

There is no transport. The file plays for as long as the source is selected.

Switching sources

Nothing else moves when the source does. The chain, the keying, the LFO and every slider keep their values, and the new picture arrives into the patch you already had. **Reset All** works the other way around: every parameter goes back to its default and the source selection stays where it is.

Whatever arrives is stretched to fill the frame. A 4:3 camera in a 16:9 output comes out wider than it went in, with no pillarbox. Matching the aspect is done by setting **Output Resolution**, which is chapter 13.

Canvas

Three controls that shape the source as it is read, before the chain. They apply to the effected picture and to the dry side of **Dry / Effected** alike, since both of them read the source through here.

Zoom

Sets how much of the frame the source fills, 0.33 to 10.0, default 1.0. Equal travel on the slider is an equal ratio of magnification, so 0.5 and 2.0 sit the same distance either side of 1.0.

At 1.0 the source fills the frame exactly. Above 1.0 the frame holds a smaller part of the picture, magnified, and 10.0 is the middle tenth of it. Below 1.0 the picture shrinks toward the center and the space around it is filled by **Border Behavior**, and at 0.33 the source sits in the middle third of the frame on both axes.

Zoom at 0.4, 1.0, and 2.5.

03-02 Zoom at 0.4, 1.0, and 2.5.

On the **Gradient Synth** there is no picture edge to shrink, so this reads as scale. Turn it up and fewer, larger cycles of color cross the frame; turn it down and more of them pack in.

Border Behavior

Sets what fills the frame wherever a coordinate lands outside the source picture. Five entries, default **Black Border**.

Nothing lands outside the source until something puts it there. Lower **Zoom** below 1.0 and all five become visible with an empty chain. Once modules are running, the geometry is what reaches past the edge, and this is what it finds.

- **Black Border** puts black outside the picture.
- **Bleed Lines** holds the outermost row and column and draws them outward, so every edge pixel becomes a line running to the frame edge.
- **Tile Screen** repeats the picture edge to edge in both directions. Each join is a hard cut.
- **Tile Reflect** repeats it mirrored, so each copy meets the last at a fold instead of a cut.
- **Gradient Synth BG** puts the Gradient Synth's own field outside the picture, running live. It is read through the same warped coordinates as everything else, so it moves with the geometry rather than sitting still behind it.

 Black Border, Bleed Lines, Tile Screen, Tile Reflect, all at Zoom 0.5.

03-03 Black Border, Bleed Lines, Tile Screen, Tile Reflect, all at Zoom 0.5.

 Gradient Synth BG. The space outside the picture is the instrument's own field, warped along with everything else.

03-04 Gradient Synth BG. The space outside the picture is the instrument's own field, warped along with everything else.

Both tiling modes make the frame edges of the source into visible lines through the middle of the picture. That is what the next control is for.

Edge Pre-Crop (Raw Input)

A switch, off by default. Turning it on reveals **Pre-Crop Pixels**, 1 to 5, default 3.

It trims that many pixels off all four edges of the incoming picture and stretches what is left back out to fill the frame. Cameras and compressed video files frequently carry a dark, smeared, or mistimed row along one or more edges. **Bleed Lines** draws that row all the way across the frame, and both tiling modes repeat it at every join, so it turns into a grid.

 Tile Screen with Edge Pre-Crop off, then on at 3 pixels.

03-05 Tile Screen with Edge Pre-Crop off, then on at 3 pixels.

At 5 pixels on a 1280 x 720 camera you are giving up under one percent of each dimension, and the stretch back to full frame is not visible at that size.

Like **Border Behavior**, this does nothing on the **Gradient Synth**.

Starting points

See all five borders. Every slot Empty, **Live Camera**, **Zoom** to 0.5. Step through **Border Behavior** and watch the space around the shrunken picture change. Now put **Zoom** back to 1.0, drop Rotation 1 into slot 1 and take its **VCO Amplitude** to about 0.6. Step through the five again. Same five behaviors, reached by the twist pulling coordinates off the edge instead of by the picture shrinking.

Build a grid, then clean it. **Live Camera**, **Border Behavior** to **Tile Screen**, **Zoom** to 0.55. The frame fills with copies and the lines between them are the camera's own edge rows. Turn **Edge Pre-Crop (Raw Input)** on and take **Pre-Crop Pixels** up until they go. Switch to **Tile Reflect** and the copies join as folds instead.

Scale the gradient. **Gradient Synth**, every slot Empty, and sweep **Zoom** across its full travel. This is the whole range of the pattern by itself.

4. Geometry Order Routing

*Before you start: **Reset All**, so the chain is back at its launch order. Any source. This chapter is about the five slots and nothing else, so leave the module sections alone until it says otherwise.*

Chapter 2 put the chain in the middle of the signal path. This is how you build one.

Five slots

The chain is five positions, numbered 1 to 5, each holding one geometry module or Empty. There are five modules to place: Mirror 1, Mirror 2, Rotation 1, Rotation 2, and Displacement.

Each slot is a menu. There is no dragging.

The line above them states the direction:

```
Signal order: slot 1 acts on the source, slot 5 is seen last
```

Under them is a readback of what you have built:

```
Source → M2 → M1 → Rot 2 → Rot 1 → Output
```

Empty slots are left out of it, so a chain of two modules reads as two modules no matter which slots they sit in. Anything you have not placed is named on the line below:

```
Not in chain: Displacement
```

Empty every slot and the readback reads `Source passes through unwarped`.

Order is the instrument

Two modules in two slots give two different pictures depending on which is which.

A mirror in slot 1 folds the source, and everything after it takes that fold and bends it. The seam ends up smeared, stretched, and carried around by whatever follows. The same mirror in slot 5 folds the finished picture instead, and the seam sits on top as a clean straight edge with the twisted image on both sides of it.

 Mirror 1 in slot 1, slot 3, and slot 5. Same settings each time.

04-01 Mirror 1 in slot 1, slot 3, and slot 5. Same settings each time.

Nothing about the module changed. Both mirrors are still at the same amplitude, the same frequency, the same angle.

This is the reason to build a patch here first and dial the modules afterwards. Sweeping a slider inside a module explores one picture. Moving that module up or down a slot gives you a different one to sweep.

Empty

An Empty slot passes the picture through untouched. It is a real bypass, not a module set to zero, and it costs nothing.

Slots do not have to be filled in order. Empty, Empty, Mirror 1, Empty, Rotation 1 behaves identically to Mirror 1 in slot 1 and Rotation 1 in slot 2. The readback confirms it: both read `Source → M1 → Rot 1 → Output`.

A module left out of the chain still has all its controls live, and none of them do anything to the picture. This is the first thing to check when a slider appears to be dead. The **Not in chain** line names exactly which modules are in that state.

A module can be in more than one slot

Listing the same module twice runs it twice, in both positions, with one set of controls driving both.

Two copies of a rotation compound into a much harder twist than the slider alone reaches. Two copies of a mirror fold the frame twice at the same angle, which is not the same as one fold, because the second one acts on the result of the first.

 Rotation 1 in one slot, two slots, and three. One slider driving all of them.

04-02 Rotation 1 in one slot, two slots, and three. One slider driving all of them.

A doubled mirror doubles in each slot it occupies, so Mirror 1 with **Doubled** on, in two slots, is four folds. See chapter 6.

Presets

Eight buttons in two rows. They set chain order and nothing else: no module parameter is touched, so the picture you get is whatever your current settings look like in that arrangement.

Each button sets all five slots at once, so nothing is inherited from what was there before.

The top row is four chains:

- **Mir→Rot** puts the mirrors on the source and the rotations after them. This is the launch order.
- **Rot→Mir** reverses that: the rotations act on the source and the mirrors fold the twisted result.
- **M2→Rot→M1** puts both rotations between the two mirrors.
- **Disp last** is the only preset that uses all five slots, with Displacement in slot 5.

The bottom row is starting points:

- **Disp 1st** is the other five-module chain, with Displacement in slot 1.
- **M1 only**, **Rot1 only**, and **Disp only** clear the chain down to one module in slot 1.

The first three leave slot 1 Empty and leave Displacement out. The three single-module buttons are the fastest way to get back to a picture you can read, and most of the starting points in this manual begin from one of them.

Luma Source Staging

Sets which picture the modules' luma controls read, 0 to 1, default 0. The caption under it reads:

0 = modules read the source video, 1 = each reads the chain so far

The luma controls are **Luma → Pivot** on the mirrors, **Luma → Twist Modulator** on the rotations, and **Luma → Displace X** and **Luma → Displace Y** on Displacement. Each one lets the brightness of a picture push its module's geometry around. This slider is what decides which picture that is.

At 0, all of them read the source, untouched, however deep in the chain they sit. Turn it up and each module reads the composite instead: the picture the rest of the chain has already warped. Its geometry starts tracking warped content rather than the source. It is a blend, not a switch, and the middle of the travel mixes the two.



Luma Source Staging at 0, then 1. The fold stops following the source and starts following the twisted picture.

04-03 Luma Source Staging at 0, then 1. The fold stops following the source and starts following the twisted picture.

Three things to know before you go looking for its effect.

It does nothing unless a luma control is turned up somewhere. With every luma control at 0 this slider has no destination and the picture does not change at any setting.

It does nothing with one module in the chain, because there is no other module for that one to read.

The module in slot 5 reads the source at any setting. The one in slot 1 reads the most. So the same two-module patch responds to this slider or does not, depending on which of the two carries the luma control.

Only the chain is included. The keyers, the hole cutter, the mix, and the vignette all run afterwards, and nothing they do reaches back into a module's geometry.

Starting points

Walk one module up the chain. Press **Disp last**, then take Rotation 1's **VCO Amplitude** to about 0.5 and Mirror 1's to about 0.35. Now move Mirror 1 from slot 2 to slot 3, then 4, then 5, setting the slot it left to Empty each time. Five arrangements of the same five modules. The seam goes from buried to sitting on top of everything.

Two of one module. Press **Rot1 only**, take **VCO Amplitude** to about 0.4, then set slot 2 to Rotation 1 as well. The readback reads Source → Rot 1 → Rot 1 → Output . Add a third copy in slot 3. One slider is now driving three twists at once and reaching a picture it cannot reach alone.

Give the staging something to do. Press **M1 only**, then set slot 5 to Rotation 1 with **VCO Amplitude** at about 0.6. Take Mirror 1's **Luma → Pivot** up to about 0.8, so the fold is following the bright parts of the picture. Now sweep **Luma Source Staging** from 0 to 1 and the fold stops following the source and starts following the twisted picture. Swap the two modules between the slots and the same sweep does nothing.

5. Oscillators

Mirrors, Rotation and Displacement are all driven by oscillators, and the controls that shape them are the same few in every module. They are described here once.

*Before you start: press **M1 only** in **Geometry Order Routing**, set the source to **Gradient Synth**, and take Mirror 1's **VCO Amplitude** to about 0.6. That puts one wave on screen in a form you can read. The oscillator controls in the other modules sit in their own sections and behave as described here except where this chapter says they differ.*

The VCO block

Every geometry module carries a group of oscillator controls at the top of its section: **Oscillator Type**, **Oscillator Mode**, **VCO Amplitude**, **VCO Frequency**, and, on some modules, a rate. That group is the VCO block.

The oscillator is a shape laid across the picture rather than a signal running in time. Move across the frame and you move along the wave. Where the wave is at a peak the module is doing the most to the picture, and where it crosses zero the module is back at rest. A still frame shows you the waveform.

The block is not identical in every module. Ranges differ, defaults differ, one family has no rate control at all, and Displacement carries a second copy of most of it. Each section below says which module has what.

Oscillator Type

One segmented picker with five entries, in this order: **Sine**, **Tri**, **Saw**, **Sqr**, **S&H**. Every module offers the same five. What changes between them is how the wave gets from one extreme to the other.

Sine curves. **Tri** runs in straight lines with a corner at each end of the travel. **Saw** climbs and then drops back in one step, so there is one hard break per cycle. **Sqr** has two values and nothing between them, so the picture holds and then jumps. **S&H** holds a random value, jumps to another, and holds again.



Sine, Tri, Saw, and Sqr on the same fold.

05-01 Sine, Tri, Saw, and Sqr on the same fold.

On a mirror the waveform is drawn along the seam and can be read off a still frame. On a rotation or on Displacement it is the pattern of how hard the picture is being pushed at each point, which reads better in motion than in a still.

S&H and S&H Slew / Lag Time

S&H steps more often than the other four cycle. At the same **VCO Frequency** you get about six steps for every cycle a sine completes, so a mirror at 5 shows about five blocks across the frame.

The values are random and they are fixed to position rather than rolled fresh. Come back to the same settings and the same blocks come back with them. Scrolling the wave walks new values into the frame instead of shuffling the ones already there.

S&H Slew / Lag Time appears under the picker when **S&H** is selected and nowhere else, 0 to 1, default 0. It ramps the start of each step toward the value being held. At 0 the steps are square. At 0.5 half of each step is a ramp and half is flat. At 1 there is no flat part left and the wave slews continuously from one value to the next.

 S&H Slew / Lag Time at 0, 0.5, and 1.

05-02 S&H Slew / Lag Time at 0, 0.5, and 1.

Oscillator Mode

Two entries, **Normal** and **Radial**. This sets the wave argument: what the wave is read along. It does not change the waveform, and the type picker works the same way in either mode.

Normal reads the wave along one axis of the frame, so the pattern comes out as parallel bands. **Radial** reads it as distance from the module's center, so the same pattern comes out as concentric rings.

 Oscillator Mode Normal, then Radial. Bands become rings.

05-03 Oscillator Mode Normal, then Radial. Bands become rings.

Rotation 1 and Rotation 2 ship in **Radial**. Mirror 1, Mirror 2 and Displacement ship in **Normal**.

What the wave is read along in **Normal** depends on the module.

On a mirror it is read along the fold line, which is what puts the ripple in the seam. In **Radial** it is measured out from the mirror's own center, so **Center X** and **Center Y** carry the rings with them.

On the rotations it is read down the frame, and **Center X** and **Center Y** do not move it. In **Radial** it is measured out from the rotation's center, which both the centers and **Dynamic Orbit Depth** move.

On Displacement one **Oscillator Mode** covers both axes. In **Normal** the X oscillator is read down the frame and the Y oscillator across it, so each axis varies in the direction it is not pushing. In **Radial** both are read from the center and land on the same rings.

Oscillator Mode on Displacement is not **Radial Push Direction**. The mode sets what the wave is read along. **Radial Push Direction** sets which way the push points. See chapter 8.

VCO Amplitude

How far the module moves the picture at the wave's peak. It is bipolar in every module, and the negative half inverts the wave: a sine slides half a cycle, a saw reverses its ramp, and a rotation twists the other way.

- Mirror 1 and Mirror 2: -1 to 1, default 0. The response is steep, so most of the deflection arrives in the last part of the travel.
- Rotation 1: -3 to 3, default 0.5, which is the twist you get before touching anything. The number is radians of twist at the wave's peak, so 3 is a little under half a turn.
- Rotation 2: -3 to 3, default 0.
- Displacement: -0.25 to 0.25, one per axis, default 0. The number is a fraction of the frame, so 0.25 pushes a quarter of the way across it.

At 0, with the module's luma control also at 0, a rotation or Displacement passes the picture through untouched. A mirror at 0 still folds and leaves a straight fold line, so a mirror is taken out with its **Enabled** switch or by setting its slot to Empty.

VCO Frequency

How tightly the wave packs into the picture.

The number is not a count of cycles. In **Normal** one full cycle takes about six of it, so a module at 5 shows a little under one cycle across the frame, and a mirror at its ceiling of 12 shows about two. **Radial** reads the wave from the center out, over half that distance, so the same setting gives about half as many rings.

- Mirror 1 and Mirror 2: 0.2 to 12, default 5.
- Rotation 1: 1 to 15, default 5.
- Rotation 2 has no frequency of its own. **Frequency Offset (vs. Rotation 1)**, -15 to 15, default 0, is added to Rotation 1's **VCO Frequency**. At 0 the two rotations run at the same frequency. Move Rotation 1's and Rotation 2's moves with it, holding whatever gap the offset sets.
- Displacement: 0.5 to 60, one per axis, default 5. The travel is curved. The bottom quarter of the track covers 0.5 to about 3.5, the top quarter covers about 46 to 60, and the middle moves fast.

Rate

Not every module has one, and the three that do call it the same thing.

Phase Speed, on Mirror 1, Mirror 2 and Displacement, is a rate, -1 to 1. On a mirror it scrolls the wave along the fold line. On Displacement there is one per axis and it moves the field. The response is squared: 0.447 is one radian per second, a full cycle in about six seconds, and full travel is five, a cycle in a little over one second. Sweeping it down through zero slows the motion to a stop and then reverses it.

The mirrors default to 0.447 and Displacement to 0, so a mirror's wave is already scrolling at launch and the displacement field holds still.

The rotations have no rate control. Their wave drifts on its own at about one cycle every thirty seconds, in both modes, and there is nothing to stop it or speed it up.

Moving a rate changes the speed without jumping the pattern, so these can be swept mid take.

What each module carries

Mirror 1 and Mirror 2 carry the whole block: type, mode, amplitude, frequency, and rate.

Rotation 1 carries type, mode, amplitude, and frequency.

Rotation 2 carries type, mode, and amplitude, with a frequency offset standing in for a frequency.

Displacement carries one **Oscillator Mode** for the module, then a full set of type, amplitude, frequency, and rate for the X axis and another for the Y axis.

What the oscillators reach

A mirror's wave is also the signal the hole cutter reads as interference, so **Oscillator Type** and **Oscillator Mode** on a mirror change where the holes fall as well as what the seam looks like. See chapter 9.

The rotations feed that stage as well. **Oscillator Type** and **S&H Slew / Lag Time** shape the rotation wave the hole cutter reads. **Oscillator Mode** does not reach it, and that signal is read out from the center in either mode.

The Global LFO runs the same five waveforms, under **LFO Waveform** and **LFO 2 Waveform**, each with its own slew control. See chapter 11.

Starting points

Read the waveform. Press **M1 only**, source **Gradient Synth**. **VCO Amplitude** to 0.6, **VCO Frequency** to 6, **Phase Speed** to 0 so the seam holds still. Step through the five entries in **Oscillator Type** and watch the seam take each shape. Stop on **S&H** and bring **S&H Slew / Lag Time** up slowly. The corners round off and the blocks turn into a slur.

Bands into rings. Same patch, **Oscillator Type** back to **Sine**. Set **Oscillator Mode** to **Radial** and **VCO Frequency** to 12. The ripple stops running along the seam and starts curving around the mirror's center. Move **Center X** to about 0.4 and the whole ring pattern goes with it.

The same block somewhere else. Press **Disp only**, source **Live Camera**. On the X axis set **VCO Amplitude** to 0.12 and **VCO Frequency** to 25, then the same two on the Y axis. Bring **Phase Speed** to 0.5 on the X axis only, so one axis moves and the other holds. Now switch **Oscillator Type** on the X axis to **Sqr**: the shimmer becomes bands that step, and the axis still holding shows you the shape they step into.

6. Mirrors

Mirror 1 and Mirror 2 are the two fold modules.

*Before you start: put Mirror 1 in slot 1 in **Geometry Order Routing** and leave the other four slots Empty. Pick any source. Everything in this chapter assumes one mirror running on its own unless it says otherwise.*

The fold

A fold takes a line across the frame and replaces everything on one side of it with a copy of the other side. That line is the fold line. The visible join it leaves behind is the seam.

Everything else in this chapter is about where the fold line sits, what shape it is, and whether it moves.



One fold. The right half of the frame replaces the left.

06-01 One fold. The right half of the frame replaces the left.

The fold line is not a straight edge unless you leave it one. The VCO block bends it into a wave, so the seam ripples instead of sitting flat. **VCO Amplitude** at zero gives you a hard mirror. Turn it up and the seam starts to breathe.

Both mirrors are identical modules with identical controls. Mirror 2 ships with its **Mirror Angle** ninety degrees off Mirror 1, which is the only difference between them out of the box.

Enabled

The switch at the top right bypasses the module wherever it sits in the chain. The slot stays where you put it and the image passes through untouched.

Mirrors are the only geometry modules with an on/off switch. Rotation and Displacement are bypassed by taking their **VCO Amplitude** to zero. **VCO Amplitude** at zero does not bypass a mirror, it leaves a straight fold.

A bypassed mirror also stops contributing to the interference pattern that cuts holes.

The VCO block

The oscillator controls work the way they do everywhere in the instrument, so chapter 5 covers the waveforms, the modes, and the S&H slew. What follows is what the oscillator does once it reaches a fold.

The oscillator output is added to the position of the fold line, measured across the fold rather than along it. Read down the seam and you are reading the waveform. A square wave gives you a stepped seam. A saw gives you a ramp that snaps back. S&H gives you a seam broken into flat blocks, and **S&H Slew / Lag Time** rounds the corners between them.

The same fold with four oscillator types. Sine, Saw, Square, S&H.

06-02 The same fold with four oscillator types. Sine, Saw, Square, S&H.

Normal reads the wave along the frame, so the ripple runs down the seam in bands. **Radial** reads it out from the module center, so the wave becomes concentric rings and the seam picks up whatever ring it crosses. Radial changes what the wave is read along. It does not change what the wave does.

VCO Amplitude

How far the seam swings, -1 to 1, default 0.

At 0 the seam is straight and the module is a plain mirror. The response is steep: half travel gives about an eighth of full deflection, and the last quarter of the slider moves fast.

Negative values invert the waveform. On a sine that reads as a half-cycle phase shift and looks like the seam has slid sideways. On a saw it reverses the ramp.

VCO Frequency

How tightly the wave packs into the seam, 0.2 to 12, default 5. The number is not a count of cycles: at the default the fold carries a little under one cycle, and at the ceiling of 12 it carries about two.

At the bottom of the range the fold holds a fraction of a cycle, so the seam is a single slow bend rather than a ripple. At the top, with **S&H** selected, the steps come close enough together to read as texture instead of shape.

At high frequency a small move of **Center X** or **Center Y** drags a lot of wave past the fold line, so the seam appears to scroll rather than shift.

Phase Speed

A rate, -1 to 1, default 0.447.

It sets how fast the wave scrolls along the fold line. Changing it mid performance changes the speed without jumping the pattern. Sweeping it across zero slows the motion to a stop and reverses it.

The response is squared. The default is one radian per second and full travel is five.



Phase Speed scrolls the wave along the seam.

06-03 Phase Speed scrolls the wave along the seam.

Luma → Pivot

Lets the brightness of the image move the fold line, -1.5 to 1.5, default 0, with a detent at zero.

The modulator is centered on mid grey. Pixels brighter than mid grey push the seam one way, darker pixels push it the other, and the seam wobbles around wherever **Mirror Angle** and the centers put it. Full deflection is about three quarters of the frame.

Because the source varies per pixel, this does not move the seam as one piece. It makes the seam follow the image content, which on a camera feed means the fold line tracks whatever is bright in the room.

Which image it reads is set by **Luma Source Staging** in Geometry Order Routing. At 0 it reads the untouched source. At 1 it reads what the chain has already built. With only one geometry module in the chain those are the same image, so the control has no effect until a second module is running.

Placement

Center X and Center Y

Where the module's anchor sits, -1 to 1 on each axis, default 0, both with a detent at zero.

The fold line passes through this point and rotates around it, so moving the center moves the fold and the reflected half with it. In **Radial** mode it is also the point the rings expand from.

Past roughly 0.6 on either axis at **Zoom 1** the seam leaves the frame. Whatever is left on screen stops being folded, so you get an unfolded image from a module that is still running.



Center X at 0, 0.35, and 0.7. Past the third the seam has left the frame.

06-04 Center X at 0, 0.35, and 0.7. Past the third the seam has left the frame.

Mirror Angle

Which way the fold line lies, 0 to 2π , default 0 on Mirror 1 and $\pi/2$ on Mirror 2.

At Mirror 1's default the fold line is vertical and the left half of the frame is replaced by the right. At Mirror 2's default it is horizontal. Running both at their defaults folds the frame into quadrants.

Spin

Turning **Spin** on replaces **Mirror Angle** with **Spin Speed (Bipolar)**, which sweeps the angle continuously instead of holding it.

The sweep picks up from wherever **Mirror Angle** was sitting, so a mirror you have placed by hand starts turning from that placement. Turning **Spin** back off returns the fold to the **Mirror Angle** slider's value.

Speed is -1 to 1, default 0, squared like the phase control. Full travel is about one complete turn every three seconds. Negative spins the other way.

A spinning fold line and a scrolling wave are two separate motions and can run at the same time.

Doubled

Doubled puts a second fold inside the same module. One module slot, two folds.

The partner fold is flat. It has no oscillator of its own: no ripple, no luma offset, no waveform. It is a straight reflection line through the module center, and it runs before the rippled fold.

The flat fold makes the frame symmetric, and the rippled seam then cuts that symmetric field, which puts the same wiggle on both sides at once.



Doubled off, then on. One seam becomes a mirrored pair.

06-05 Doubled off, then on. One seam becomes a mirrored pair.

Radial has nothing to read on the partner fold, so in **Doubled** the partner behaves the same in either mode.

Doubled Angle Offset

How far the partner's line is turned away from the base fold's line, 0 to 2π , default π .

At the default the two lines are opposed, which is the full frame mirror above. This slider has no zero detent: the home value is π , and the per-slider reset button returns to π .

Turn it away from π and the two fold lines stop being parallel. The pair becomes a wedge, and the image inside the wedge repeats around the module center the way a folded corner does.

 Doubled Angle Offset at 2.1. The two fold lines stop being parallel and the pair becomes a wedge.

06–06 Doubled Angle Offset at 2.1. The two fold lines stop being parallel and the pair becomes a wedge.

A doubled module doubles in every slot it occupies. Two slots is four folds.

Running both mirrors

Two mirrors compound. Two folds at ninety degrees fold the frame into quadrants, and a third fold from a doubled module or a repeated slot is close to saturation: past that point most of the frame is a copy of a copy and further folds stop adding visible structure.

Where they sit relative to each other in the chain matters. A mirror in slot 1 acts on the source, and a mirror in slot 2 acts on what slot 1 produced. Swapping the two gives a different picture from the same settings.

The two mirrors also produce the interference pattern that the hole cutter reads, which is what chapter 9 works on. **Wave Folding** does nothing in a chain with no mirrors in it, because there is no interference to fold.

What the mirrors feed

Keyer 1 — Mirrors taps this family. Its Feed picker chooses between the mirrors' own output and the finished warp, which is what lets you key on the folds rather than on the composite. See chapter 10.

Two Global LFO destinations land here. **LFO → Mirror Amp** drives both modules' **VCO Amplitude** as a pair, spread apart by **Bus Spread**. **LFO → Mirror Center** drives both fold centers as an XY path, with Mirror 2's axes swapped so the two centers travel oppositely. See chapter 11.

Starting points

One mirror, one motion. Mirror 1 in slot 1, everything else Empty. **VCO Amplitude** to about 0.35, **VCO Frequency** to 3. Sweep **Mirror Angle** slowly through a full turn and watch the seam walk around the frame. Add **Spin** at a low speed and leave it running.

The doubled wedge. Same patch, turn **Doubled** on and set **Doubled Angle Offset** to about 2.1. Bring **VCO Frequency** up to 8. The wedge repeats the ripple around the center. Move **Center X** and **Center Y** off zero and the figure slides while staying folded.

Content driven. Live Camera, one mirror, **VCO Amplitude** at 0, so fold is straight. Bring **Luma → Pivot** up to about 0.8. The fold line stops being a line and follows whatever is bright in the frame. Put Rotation 1 in slot 2 with a small **VCO Amplitude** and **Luma Source Staging** starts to have something to do.

7. Rotation

Rotation 1 and Rotation 2 are the two twist modules.

*Before you start: press **Rot1 only** in **Geometry Order Routing**, which puts Rotation 1 in slot 1 and clears the rest. Pick any source. Everything in this chapter assumes one rotation running on its own unless it says otherwise.*

The twist

A twist turns the picture about a center point. Every pixel moves around that center along its own circle, and none of them change how far out they sit. The middle of the frame barely moves and the outside sweeps a long way, so straight lines bend into arcs and the picture winds up.

How far each point turns is set by the oscillator, read at that point. At **Radial**, which is how both rotations ship, the wave is read as distance from the center, so the frame turns by different amounts at different radii and the result is a spiral.

 One rotation. Straight lines bend into arcs and nothing leaves its ring.

07-01 One rotation. Straight lines bend into arcs and nothing leaves its ring.

Nothing is folded and nothing is duplicated, so a rotation leaves no seam. At high amplitude the corners of the picture swing outside the frame and whatever **Border Behavior** is set to shows up along the edges.

Both rotations are twist modules with the same core. Rotation 1 carries a frequency and a luma control that Rotation 2 does not, and Rotation 2 carries three orbit controls that Rotation 1 does not.

Taking a rotation out

Rotation has no on/off switch. Set **VCO Amplitude** to 0, with **Luma → Twist Modulator** also at 0, and the module passes the picture through untouched wherever it sits. Setting its slot to Empty does the same thing.

Rotation 1 ships at **VCO Amplitude** 0.5, so there is twist on screen at launch before anything is touched.

The VCO block

The oscillator controls work the way they do everywhere in the instrument, so chapter 5 covers the waveforms, the modes, and the S&H slew. What follows is what the oscillator does once it reaches a twist.

The wave sets the amount of turn at each point. Where it peaks, the picture turns hardest. Where it crosses zero, that part of the picture sits where it started. In **Radial** those zero crossings are circles, and they read as still rings with the picture shearing between them. In **Normal** the wave is read down the frame instead, so the amount of turn varies from top to bottom and the rings become horizontal bands of shear.

The rotations have no rate control, and their wave drifts on its own at about one cycle every thirty seconds. In **Radial** that shows as the rings slowly walking outward.

VCO Amplitude

How far the picture turns at the wave's peak, -3 to 3, default 0.5 on Rotation 1 and 0 on Rotation 2.

The number is radians, so 3 is a little under half a turn and 1.57 is a quarter turn. Below about 0.3 the picture reads as bent rather than twisted. Above 2 the outer part of the frame wraps past itself and the picture starts tearing into strands.

Negative values twist the other way. Sweeping through zero unwinds the picture and rewinds it in the opposite direction.



VCO Amplitude at 0.5, 1.5, and 3.

07-02 VCO Amplitude at 0.5, 1.5, and 3.

VCO Frequency

How many rings of twist fit into the picture, 1 to 15, default 5. This control is on Rotation 1 only, and Rotation 2's frequency is derived from it.

At the bottom of the range the whole frame turns almost as one piece, with the shear spread out over the entire radius. At the top the rings are close enough together that neighboring bands turn opposite ways, which is where the strand and ribbon look comes from.

Luma → Twist Modulator

Lets the brightness of the image set part of the twist, -0.30 to 0.30, default 0, with a detent at zero.

The modulator is centered on mid grey. Pixels brighter than mid grey twist one way, darker pixels twist the other, and the amount adds to whatever **VCO Amplitude** is set to. At full travel it reaches as far in either direction as the whole **VCO Amplitude** slider does, so a bright frame at 0.30 twists as hard as amplitude 3 on its own.

Because the source varies per pixel, this does not turn the picture as one piece. Bright and dark areas turn by different amounts, which smears the image along its own contours.

Which image it reads is set by **Luma Source Staging** in Geometry Order Routing. At 0 it reads the untouched source. At 1 it reads what the chain has already built. With only one geometry module in the chain those are the same image, so the control has no effect until a second module is running.

Rotation 2 has no luma control.

Placement

Center X and Center Y

Where the module turns about, -1 to 1 on each axis, default 0. Each rotation has its own pair.

Moving the center moves the eye of the spiral, and everything winds around the new position. In **Radial** the rings move with it, because they are measured from this point. In **Normal** the wave stays where it is and only the pivot moves, so the picture turns about a new place through the same bands of shear.

Past about 0.5 on either axis the center leaves the frame. What is left on screen is one arm of the spiral sweeping through, which turns the module into a directional smear rather than a twist.

Dynamic Orbit Depth

Puts the center on a moving path, 0 to 1, default 0.

The path is added to **Center X** and **Center Y** rather than replacing them, so the sliders still set where the figure sits and this control sets how far it wanders from there. At 1 the center travels about a third of the frame width either side of that home position.

The path is driven by the mirrors' oscillators. On Rotation 1, the horizontal part of the travel follows Mirror 1 and the vertical part follows Mirror 2. Two things fall out of that, and both are visible with a single rotation on screen:

- The speed of the orbit is set by the mirrors' **Phase Speed** controls. With both at 0 the center stops somewhere along the path and stays there, which is a way to place the center outside the range of the **Center X** and **Center Y** sliders.
- The size of the orbit grows with the mirrors' **VCO Amplitude**, from about a third of the frame width at 0 up to about half the frame width at full.

Neither mirror has to be in the chain, and neither has to be **Enabled**. The oscillators run regardless, and this control reads them wherever they are.

 Dynamic Orbit Depth at 1. The eye of the spiral circles, driven by two mirrors that are not in the chain.

07-03 Dynamic Orbit Depth at 1. The eye of the spiral circles, driven by two mirrors that are not in the chain.

Rotation 2

Rotation 2 is a second twist with its own oscillator, its own center, and its own amplitude. Everything above applies to it, with the differences below.

Frequency Offset (vs. Rotation 1)

Sets Rotation 2's frequency as a distance from Rotation 1's, -15 to 15, default 0.

At 0 the two rotations run at the same frequency and their rings sit on top of each other. Move Rotation 1's **VCO Frequency** and Rotation 2's frequency moves with it, holding whatever gap this control sets. Small offsets put the two ring patterns slightly out of step so they beat against each other, and the beat is what this control plays.

Taking the offset negative far enough cancels Rotation 1's frequency. At that point Rotation 2 has no rings at all: the whole frame turns as one rigid piece, swinging slowly back and forth as the wave drifts. Rotation 1 cannot reach that state, because its frequency stops at 1.

 Rotation 2 at Frequency Offset 0, 3, and -5. At -5 the rings are gone and the whole frame turns as one piece.

07-04 Rotation 2 at Frequency Offset 0, 3, and -5. At -5 the rings are gone and the whole frame turns as one piece.

Dynamic Orbit Depth

Same control, 0 to 1, default 0, and the path is built the other way round. Rotation 2's horizontal travel follows Mirror 2 and its vertical travel follows Mirror 1, so the two rotations' centers do not trace the same figure and do not stay in line with each other.

Orbit Phase Offset

Where Rotation 2 sits along its path relative to Rotation 1, 0 to 1 for a full turn, default 0.

At 0 the two centers set off together. At 0.5 they sit on opposite sides of their paths and pass each other in the middle. This does nothing while **Dynamic Orbit Depth** is at 0.

Orbit Lissajous Ratio

The shape of Rotation 2's path, 0.25 to 4, default 1.

It stretches the horizontal part of the travel against the vertical, which is what changes the figure rather than its size. At 1 the center traces an ellipse. At 2 it traces a figure eight. At 3 it traces a three-lobed path. Between those values the figure does not close, so the center keeps covering new ground instead of repeating.

The two mirrors' **Phase Speed** settings feed the two axes, so setting them to different rates changes the figure as well. This also does nothing while **Dynamic Orbit Depth** is at 0.

Running both rotations

Two rotations compound. Their twist angles add up where their waves agree and cancel where they oppose, so two modules at moderate amplitude reach further than either does alone and also produce still rings that neither has by itself.

Where they sit relative to each other in the chain matters. A rotation in slot 1 twists the source, and a rotation in slot 2 twists that result. Swapping them gives a different picture from the same settings, because the second one turns about its center in a frame that is already wound up.

The same rotation listed in two slots runs twice, from one set of controls. Two copies at amplitude 1 reach further than one copy at 2, because the second pass twists a picture whose radii have already been rearranged.

What the rotations feed

Keyer 2 — Rotation taps this family. Its Feed picker chooses between the rotations' own output and the finished warp. The family tap always runs Rotation 1 and then Rotation 2, whatever order the chain has them in. See chapter 10.

The hole cutter reads a rotation wave of its own, built from both modules' **Oscillator Type**, **S&H Slew / Lag Time**, and frequency. **Rotation 1 + 2 Combine**, in the **Keying** section, sets how the two are combined into it: Additive, Subtractive, XOR, or Multiply. Multiply outputs nothing until Rotation 2 has some amplitude. How much of the result reaches the holes is set by **Rotation → Holes** directly above it. See chapter 9.

That wave is read out from the center whichever way **Oscillator Mode** is set, and its rings are twice as tight as the twist rings on screen.

Two Global LFO destinations land here. **LFO → Rotation Amp** drives both modules' **VCO Amplitude** as a pair, spread apart by **Bus Spread**. **LFO → Rotation Center** drives both centers as an XY path, with Rotation 2's axes swapped so the two centers travel oppositely. See chapter 11.

Starting points

One spiral, read the rings. Press **Rot1 only**. Take **VCO Amplitude** to about 1.2 and **VCO Frequency** to 8. The still rings are where the wave crosses zero, and the shear between them is where it peaks. Now switch **Oscillator Mode** to **Normal**: the same amount of twist arrives as horizontal bands instead, and the figure stops being centered on anything.

The rigid turn. Press **Rot1 only**, then set slot 2 to Rotation 2. Leave Rotation 1's **VCO Frequency** at 5 and set Rotation 2's **Frequency Offset (vs. Rotation 1)** to -5. Bring Rotation 2's **VCO Amplitude** up to about 1.5. Rotation 2 now turns the whole frame as one piece, slowly swinging one way and then the other, while Rotation 1 keeps spiralling underneath it.

Drive the orbit from a module that is not in the chain. Press **Rot1 only**, **VCO Amplitude** to about 1.0, then **Dynamic Orbit Depth** to 1. The eye of the spiral starts circling. Go to Mirror 1 and move its **Phase Speed**: the orbit changes speed on one axis. Bring Mirror 1's **VCO Amplitude** up and the orbit widens. Neither mirror is in the chain.

8. Displacement

Displacement is the fifth geometry module, and the only one that keeps nothing about where a pixel started.

*Before you start: press **Disp only** in **Geometry Order Routing**, which puts Displacement in slot 1 and clears the rest. Pick a source with detail in it, because this module needs something to move. Both amplitudes ship at 0, so the picture will not change until you turn one of them up.*

The push

A push moves each pixel by an amount, in a direction, and lets it land wherever that puts it. A fold leaves a pixel either untouched or exactly reflected. A twist keeps every pixel at the same distance from a center. Displacement holds neither of those, and that is what makes it read as liquid rather than as geometry: heat shimmer, water, a bad tape.

There is no seam and no still ring. What you get instead is the picture sliding against itself.



Both axes pushing. Rows slide against each other sideways and columns slide vertically at the same time.

08-01 Both axes pushing. Rows slide against each other sideways and columns slide vertically at the same time.

The module runs two oscillators, one per axis, and they are crossed. The X oscillator is read down the frame, so how far the picture is pushed sideways depends on how far down the screen you look. The Y oscillator is read across the frame, so how far it is pushed up or down depends on how far across you look. Rows slide against each other horizontally and columns slide against each other vertically, at the same time.

At the frame edges the push reaches for picture that is not there, so whatever **Border Behavior** is set to shows up along the sides. On the **Gradient Synth** it never runs out of picture and no border appears at all.

This is not rotation

At **Radial Push Direction** 1 the Y oscillator pushes each pixel sideways around the center, which looks like a swirl and is not one.

A rotation turns every pixel about the center by an angle, so each one stays on its own circle and the picture winds up without changing what is at any given radius. Displacement moves each pixel a fixed distance along the direction its tangent points, so pixels leave their circles: some rings pull apart, others pile into each other, and the

picture smears. Two settings that look similar in a still behave differently the moment either one moves. See chapter 7.

Taking it out

Displacement has no on/off switch. With both axes at **VCO Amplitude 0** and both **Luma → Displace** controls at 0, the picture passes through untouched wherever the module sits. Setting its slot to Empty does the same thing.

The two axes

Under the module's own controls the section splits into **X Axis** and **Y Axis**, and each block holds a full set: **Oscillator Type**, **S&H Slew / Lag Time**, **VCO Amplitude**, **VCO Frequency**, **Phase Speed**, and a luma control. The two blocks are independent, so the horizontal and vertical push can run at different sizes, different densities, different waveforms, and different speeds.

Everything above the two blocks is one control for the whole module: **Oscillator Mode**, **Radial Push Direction**, **Center X**, and **Center Y**.

The oscillator controls work the way they do everywhere in the instrument, so chapter 5 covers the waveforms, the modes, and the S&H slew.

VCO Amplitude

How far the picture is pushed at the wave's peak, -0.25 to 0.25 on each axis, default 0.

The number is a fraction of the frame, so 0.25 moves a pixel a quarter of the way across it. Negative reverses which way the peak pushes.

One axis on its own gives sliding rows or sliding columns. Both at the same value gives a field that pushes on the diagonal. Both up with different frequencies is where the picture stops looking pushed and starts looking poured.

VCO Frequency

How fine the field is, 0.5 to 60 on each axis, default 5.

This is the widest frequency range in the instrument and its travel is curved. The bottom quarter of the track covers 0.5 to about 3.5, the top quarter covers about 46 to 60, and the middle moves fast. Low settings give one

broad swell across the whole frame, which reads as the picture leaning. High settings give a fine grain that reads as texture and, on a camera, as noise with structure in it.

 VCO Frequency at 2, 10, and 45, both axes.

08-02 VCO Frequency at 2, 10, and 45, both axes.

Phase Speed

The rate control, -1 to 1 on each axis, default 0. The response is squared: one radian per second at 0.447, five at full travel, and sweeping down through zero stops the motion and reverses it.

At the default the field holds still, so Displacement at rest is a fixed distortion rather than a moving one. That is the state to dial an amplitude in at, and one axis can be left frozen while the other runs.

Luma → Displace X and Luma → Displace Y

Let the brightness of the image push the picture, -0.30 to 0.30 on each axis, default 0, with a detent at zero.

This does not sit around mid grey the way the mirrors' and the rotations' luma controls do. Black pushes nothing and white pushes the full amount, so turning it up shifts the bright parts of the picture in one direction while the dark parts stay where they are. Bright areas drag and dark areas hold.

At full travel it reaches further than the **VCO Amplitude** slider above it, so a bright frame at 0.30 displaces further than the oscillator can on its own.

 Luma → Displace X at 0, then 0.25. Bright areas slide and dark areas hold.

08-03 Luma → Displace X at 0, then 0.25. Bright areas slide and dark areas hold.

Which image it reads is set by **Luma Source Staging** in Geometry Order Routing. At 0 it reads the untouched source. At 1 it reads what the chain has already built. With only one geometry module in the chain those are the same image, so the control has no effect until a second module is running.

Oscillator Mode

One picker for both axes. **Normal** is the crossed cartesian field described above. **Radial** reads both oscillators as distance from the module center, so the field collapses onto concentric rings and the picture moves like water around a dropped stone.

With both axes at the same **VCO Frequency** and the same **Oscillator Type**, **Radial** gives both oscillators the same wave to read, so the push points one fixed way and only its strength rings. Detune the two frequencies and the direction starts turning with the radius as well.

Radial Push Direction

Which way the two oscillators point their push, 0 to 1, default 0. The line under it reads:

At 0, X pushes horizontally and Y vertically. At 1, X pushes outward from center and Y swirls around it.

At 0 the X oscillator drives left and right and the Y oscillator drives up and down. At 1 the X oscillator drives in and out from **Center X** and **Center Y**, so the picture breathes, and the Y oscillator drives around that point, so it swirls. Running both at 1 gives breathing and swirling together, which is the spiral smear.

It is a blend, not a switch. The middle of the travel is a real field with the push pointing part way between the two, and it is the only place a slow sweep of the control has anything to sweep through.



Radial Push Direction at 0, 0.5, and 1.

08-04 Radial Push Direction at 0, 0.5, and 1.

Radial Push Direction is not **Oscillator Mode**. The mode sets what the wave is read along. This sets which way the result points. Either one can be set without the other, and the four combinations are four different fields.

Center X and Center Y

Where the module is anchored, -1 to 1 on each axis, default 0. One pair for the whole module.

It moves the wave pattern and the radial basis together, so the field and the point it breathes around stay attached to each other rather than drifting apart. With **Radial Push Direction** at 0 the center still moves the pattern, and the push keeps pointing along the frame's own axes.

At high **VCO Frequency** a small move of the center drags a lot of wave past any given point, so the picture appears to scroll rather than shift. That is the same behavior the mirrors show at high ripple frequency.

What Displacement feeds

Keyer 3 — Displacement taps this module. Its Feed picker chooses between the displaced picture and the finished warp. See chapter 10.

In a chain with no mirror in it and **Rotation → Holes** at 0, the hole cutter reads how hard Displacement is pushing at each point instead of mirror interference. The holes then follow the field, open where the push is strongest and closed where it is weakest, so **Collision Threshold** has something to work against. **Wave Folding** does not: it runs earlier and does nothing in a chain with no mirrors in it, whatever else is there. See chapter 9.

Three Global LFO destinations land here. **LFO → Displace Amp** drives both axes' **VCO Amplitude**, spread apart by **Bus Spread**, and at quadrature the two axes sit a quarter cycle apart and the field moves in circles rather than pulsing on a diagonal. **LFO → Displace Center** drives the module's one center as an XY path. **LFO → Radial Push** sweeps **Radial Push Direction** between cartesian and radial. See chapter 11.

Starting points

Shimmer. Press **Disp only**, source **Live Camera**. Set both axes to **VCO Amplitude** 0.08 and **VCO Frequency** 30. Bring **Phase Speed** up to about 0.4 on the X axis only. One axis moves and the other holds, so the picture crawls in one direction while staying pinned in the other. Take the X axis **Oscillator Type** to **Sqr** and the crawl breaks into sliding blocks.

Pond. Same patch, **VCO Frequency** to 20 on both axes, **Oscillator Mode** to **Radial**, **Radial Push Direction** to 1. The field turns into rings that breathe and swirl at once. Now set the Y axis **VCO Frequency** to 14 and the rings stop agreeing about which way to point.

Let the picture push itself. Press **Disp only**, source **Live Camera**, both amplitudes at 0 so the oscillators are silent. Bring **Luma → Displace X** up to about 0.2. Everything bright in the room slides sideways and everything dark stays put. Put Rotation 1 in slot 2 with a small **VCO Amplitude** and sweep **Luma Source Staging**, and the bright parts it is following become the twisted picture rather than the source.

9. Interference and holes

The first stage after the chain cuts parts of the picture out and leaves them black.

*Before you start: **Reset All**, set slot 1 to Mirror 1 and slot 2 to Mirror 2 with every other slot Empty, and turn **Enabled** on at the top of the **Keying** section. Nothing in this chapter does anything until that switch is on.*

Where this happens

The hole cutter runs on what the chain produced, before the three keyers. Its controls sit in the **Keying** section above them: **Collision Threshold**, **Wave Folding**, and **Rotation → Holes**, with **Rotation 1 + 2 Combine** beneath them.

The **Enabled** switch at the top of that section gates all of it, and it is off at launch. With it off, every control in this chapter is inert.

What gets cut goes black. **Invert Key** swaps which parts are cut and which are kept, and **Key Softness (Global)** feathers the edge, both covered in chapter 10.

The interference signal

Each mirror in the chain produces a wave, the one described in chapter 5. The hole cutter reads those waves rather than the picture.

With both mirrors running, the two waves are multiplied together. The signal is strong only where both are near a peak at the same point, and it collapses wherever either one passes through zero. That gives a grid of strong spots with quiet ground between them, and the geometry of that grid is the beat between the two mirrors' settings.

With one mirror running, the signal is that mirror's wave alone, which gives bands rather than spots. With no mirror running, there is no signal, and the stage stays silent unless one of the two substitutes further down applies.



One mirror gives bands. Two mirrors multiply into a grid.

09–01 One mirror gives bands. Two mirrors multiply into a grid.

The signal does not depend on VCO Amplitude. A mirror folding at amplitude 0 contributes exactly as much to the interference as one folding at full travel. Take both mirrors to 0 and the picture has two hard straight folds in it and a complete hole pattern on top of them.

What changes the pattern is everything else about the mirrors' oscillators: **VCO Frequency** sets how fine it is, **Oscillator Type** sets the shape of the spots, **Oscillator Mode** turns bands into rings, **Phase Speed** makes the whole pattern crawl, and **Center X**, **Center Y** and **Mirror Angle** move and turn it.

A mirror with **Enabled** off, or one that is not in the chain, contributes nothing. Two mirrors in the chain with one of them switched off gives the one-mirror signal.

Collision Threshold

How strong the signal has to be before the picture is cut, 0.05 to 0.95, default 0.7.

At 0.95 only the very strongest points are cut, so the holes are small and sparse and sit exactly on the peaks. Bring it down and each hole grows outward from its peak until the holes meet each other and it is the picture that is left in islands. At 0.05 most of the frame is gone.



Collision Threshold at 0.9, 0.6, and 0.3.

09-02 Collision Threshold at 0.9, 0.6, and 0.3.

Sweeping this control with the mirrors' **Phase Speed** running is the difference between a static stencil and a picture that opens and closes.

Darks has no effect here. That control flips the three keyers, not the hole signal.

Wave Folding

Folds the interference back on itself, 0 to 2, default 0.

At 0 each peak in the signal produces one hole. Raise it and the same peak is folded into several nested bands, so one hole becomes a set of concentric rings around where it was. The higher the setting, the more bands, and the picture ends up cut into fine contours that follow the interference rather than sitting on top of it.



Wave Folding at 0, 0.6, and 1.6.

09-03 Wave Folding at 0, 0.6, and 1.6.

Wave Folding acts on mirror interference and nothing else. In a chain with no mirror in it there is nothing to fold, and this control does nothing at any setting. That holds whatever else is running: a rotation-only or displacement-

only chain can produce holes by the routes below, and this control still will not touch them.

Rotation → Holes

Mixes the rotations' own wave into the hole signal, 0 to 1, default 0.

With a mirror in the chain, the rotation wave multiplies against the interference. Holes survive only where both signals are strong, so the existing pattern picks up concentric rings and goes quiet along the rings where the rotation wave is low. Turning the control up moves between the mirror pattern and that ringed version of it.

With no mirror in the chain, the rotation wave becomes the signal outright. Holes then follow rings measured from the rotation's center, and the whole stage works in a chain that has no mirror in it at all.



Rotation → Holes at 0, then 1, in a chain with no mirror in it.

09-04 Rotation → Holes at 0, then 1, in a chain with no mirror in it.

That wave is read out from the center whichever way **Oscillator Mode** is set, and its rings are twice as tight as the twist rings on screen. It is built from both rotations' **Oscillator Type**, **S&H Slew / Lag Time** and frequency, and **Rotation 1 + 2 Combine** directly beneath sets how the two are combined: Additive, Subtractive, XOR, or Multiply. Multiply outputs nothing until Rotation 2 has some amplitude. See chapter 7.

Holes from Displacement

In a chain with no mirror in it and **Rotation → Holes** at 0, the signal becomes how hard Displacement is pushing at each point. Holes open where the push is strongest and close where it is weakest, so they travel with the field and move when **Phase Speed** is up.

There is no control for this. It applies when those conditions are met, and **Collision Threshold** works against it the same way it works against interference. Putting a mirror back in the chain, or raising **Rotation → Holes**, hands the signal back to whichever of those has priority.

What the holes feed

The hole coverage is combined with the three keyers, folded together by OR or by XOR one keyer at a time, and the result is the shape that goes black. A hole and a keyed area are the same kind of thing by the time they reach the picture, which is why one section holds both. See chapter 10.

Everything cut here is cut before the mix and the vignette, so **Dry / Effected** blends a holed picture against an unholed one. See chapter 12.

Starting points

Holes without folds. Slot 1 Mirror 1, slot 2 Mirror 2, both at **VCO Amplitude 0**, **Enabled** on in **Keying**. The picture has two straight folds and a grid of holes cut into it. Now move Mirror 2's **VCO Frequency** from 5 up to 9 and watch the grid stretch and re-space while the folds stay exactly where they are.

Make it breathe. Same patch, bring **Collision Threshold** down to about 0.45 so the holes are large, then set both mirrors' **Phase Speed** to about 0.3. The pattern crawls. Sweep **Collision Threshold** slowly by hand between 0.35 and 0.75 and the picture opens and closes against that crawl.

Rings from nothing. Press **Rot1 only**, leave **Enabled** on, and take **Rotation → Holes** to 1. There is no mirror in the chain, so the rings are the whole signal. Set Rotation 1's **VCO Frequency** to 9 for more of them, then put Rotation 2 in slot 2 with **VCO Amplitude** at 1 and set **Rotation 1 + 2 Combine** to XOR: the two ring sets fold against each other and the holes land where they disagree.

10. Keying

The three keyers cut parts of the picture out by brightness and leave them black, the same black the holes leave.

*Before you start: **Reset All**, set slot 1 to Mirror 1 and slot 2 to Rotation 1 with every other slot Empty, set Mirror 1's **VCO Amplitude** to 0.3 and its **Phase Speed** to 0, and turn **Enabled** on at the top of the **Keying** section. The picture is a folded, twisted gradient with the hole bands from chapter 9 cut into it.*

The section and its switch

The **Enabled** switch at the top of **Keying** gates the whole stage, the hole cutter included, and is off at launch. Below it sit **Invert Key** and **Darks** on one row, then **Key Softness (Global)**, then the four hole-cutter controls covered in chapter 9, then the three keyers, each a group of three controls: a threshold slider, a Feed picker, and an XOR checkbox.



The Keying section. Two switches, the global softness, the hole cutter, and three keyers of three controls each.

10-01 The Keying section. Two switches, the global softness, the hole cutter, and three keyers of three controls each.

Turning **Enabled** on at launch shows the hole pattern and nothing else. All three keyers ship silent: **Keyer 1 — Mirrors** sits at 1.00, and a keyer whose threshold is at 0.99 or above cuts nothing. The keyers enter the picture when that slider comes down.

The three keyers

A key is a threshold decision: everywhere the signal a keyer reads is brighter than its threshold, the picture is cut to black. The three keyers are fixed to the module families. **Keyer 1 — Mirrors** reads the Mirrors, **Keyer 2 — Rotation** reads the Rotations, and **Keyer 3 — Displacement** reads Displacement, each through the Feed picker described in the next section.

Keyer 1 — Mirrors holds the one absolute threshold, 0 to 1, default 1. At 1.00 it cuts nothing. Pull it down and the cut starts in the brightest parts of its feed and grows darker-ward as the number falls.

 Keyer 1 — Mirrors at 0.75, 0.50, and 0.25.

10-02 Keyer 1 — Mirrors at 0.75, 0.50, and 0.25.

Keyer 2 — Rotation and **Keyer 3 — Displacement** are offsets from that threshold, -1 to 1, default 0, with a detent at zero. Each one's working threshold is **Keyer 1 — Mirrors** plus its own offset, held between 0 and 1. At the detent all three keyers share one threshold, so dragging **Keyer 1 — Mirrors** moves the whole group together and the offsets set the gaps inside it.

The 0.99 rule applies to the working threshold, not the slider. To silence Keyer 2 or Keyer 3 while the others stay live, push its offset up until the sum reaches 1: an offset of 1.00 does it from any threshold, and anything past the point where the sum first reaches 1 lands on the same silence. Pushed the other way, an offset that drags the sum to 0 puts the threshold at black, and the keyer covers everything in its feed that is not black. With **Darks** on, the same position covers nothing.

Feeds

Each keyer's Feed picker chooses its tap, the signal it reads. Two entries: the keyer's own family, or **Warped Final**.

Warped Final is the finished warp, the picture as the chain left it and as it appears on screen before any cutting. A keyer fed from it cuts by the brightness of what is in front of you.

The family entry (**Mirrors**, **Rotation**, or **Displacement**) reads what that family does to the source on its own: the source refolded, retwisted, or repushed by that family alone, measured for brightness. The chain has no say in it. Slot order does not change a family tap, and a module does not need to be in the chain to appear in one. The **Mirrors** tap carries every mirror whose own **Enabled** switch is on, doubled folds included, and both mirrors ship switched on, so the tap has their folds in it even when no mirror is in the chain. The **Rotation** tap always carries both rotations, and the **Displacement** tap always carries Displacement; those modules go quiet in a tap the same way they go quiet in the chain, with amplitude and their luma controls at 0.

 One keyer, two feeds. Mirrors cuts by the fold; Warped Final cuts by the picture.

10-03 One keyer, two feeds. Mirrors cuts by the fold; Warped Final cuts by the picture.

A family at rest hands its tap the plain source, so its keyer becomes a straight brightness key on the source itself. In the chapter patch Displacement is at rest, and pulling **Keyer 3 — Displacement** below the group cuts the gradient's bright regions exactly where the source has them.

Darks

Darks flips which side of the threshold every keyer cuts: on, each keyer cuts where its feed is darker than the threshold instead of brighter. It flips all three keyers at once and does not touch the hole signal, so the hole pattern stays where it was while the keyed regions trade places with their surroundings.

Key Softness (Global)

Feathers every threshold edge in the section, 0 to 0.35, default 0.

At 0 every edge is a hard line: a point is cut or it is not. Raise it and each edge becomes a fade from picture to black, and the fade widens as the number climbs, until at 0.35 the edge is a broad ramp. One slider governs all of it: the three keyers, and the hole cutter's **Collision Threshold** edge with them.



Key Softness (Global) at 0 and at 0.35.

10-04 Key Softness (Global) at 0 and at 0.35.

The fades survive everything downstream in the section. Shapes combined by OR or XOR keep their soft edges, and **Invert Key** inverts a fade into a fade.

How the shapes combine

The cut is assembled in a fixed order: the hole coverage first, then **Keyer 1 — Mirrors**, then **Keyer 2 — Rotation**, then **Keyer 3 — Displacement**.

Each keyer folds in as OR: its cut adds to the shape, and the picture is cut wherever either the shape so far or the keyer says so. Turn a keyer's **XOR** checkbox on and that keyer folds in as XOR instead: it cuts where it alone cuts, and it re-opens the picture wherever its coverage lands on ground the shape so far had already cut. With **XOR** on **Keyer 1 — Mirrors**, the holes are what it plays against, and the picture comes back inside every hole the keyer also covers.



The same keyer folded in by OR, then by XOR. XOR re-opens the holes it lands on.

10-05 The same keyer folded in by OR, then by XOR. XOR re-opens the holes it lands on.

Invert Key

Invert Key runs after all three keyers have folded in and swaps cut for kept across the whole shape, holes included.

It is not **Darks** under another name. **Darks** changes what each keyer covers before the shapes combine; **Invert Key** flips the finished result. With one keyer and no holes the two look alike. With holes in the frame they do not: **Darks** moves the keyed regions to the dark side of the picture and leaves the holes standing, while **Invert Key** turns the holes into the only places the picture survives.



Neither switch is the other. Darks moves the keyed regions; Invert Key flips the whole shape, holes included.

10-06 Neither switch is the other. Darks moves the keyed regions; Invert Key flips the whole shape, holes included.

The difference key

Two keyers at the same threshold on different feeds, with both their **XOR** checkboxes on, cut only where the two feeds disagree: where one feed is above the threshold and the other is below it. Where both feeds agree, cut against cut cancels and the picture stands.

The clean version needs the hole signal quiet, which a chain with no mirror provides. Press **Rot1 only**, leave **Enabled** on, and set **Keyer 1 — Mirrors** to 0.5 with its Feed on **Mirrors** and **XOR** on. Then leave **Keyer 2 — Rotation** at its detent, set its Feed to **Warped Final**, and turn its **XOR** on. Keyer 1 reads the source folded by the two idle mirrors; Keyer 2 reads the twisted picture on screen; both threshold at 0.5. The black lands only where the fold and the twist disagree about which side of 0.5 a point sits on, and it moves as the twist drifts against the still folds.



The difference key. Black only where the fold and the twist disagree.

10-07 The difference key. Black only where the fold and the twist disagree.

Starting points

First cut. Slot 1 Mirror 1, every other slot Empty, Mirror 1 **VCO Amplitude** 0.3 and **Phase Speed** 0.3, **Enabled** on in **Keying**. Push **Keyer 2 — Rotation** and **Keyer 3 — Displacement** to 1.00 so Keyer 1 works alone, then pull **Keyer 1 — Mirrors** down slowly from 1.00. Nothing moves until 0.99, then the bright parts of the fold start going black, and the cut crawls with the fold. Turn **Darks** on and the cut jumps to the dark side of the same picture.

Soft and breathing. Same patch at **Keyer 1 — Mirrors** 0.5. Raise **Key Softness (Global)** to 0.2: every hard edge in the frame, holes included, becomes a fade, and the crawl reads as breathing rather than switching. Sweep **Collision Threshold** between 0.5 and 0.9 and the hole bands swell and thin through their soft edges.

Where they disagree. The difference key patch above, then take Mirror 1's **VCO Amplitude** up past 0.5. The fold in Keyer 1's tap ripples harder, the disagreement with the twisted picture grows, and the black spreads without a single threshold moving.

11. Global LFO

One modulator drives the whole instrument. It moves controls rather than pixels: a value that changes over time, applied to sliders you would otherwise move by hand.

*Before you start: **Reset All**, press **M1 only**, and set Mirror 1's **VCO Amplitude** to 0.6 and its **Phase Speed** to 0. The picture holds still. Every destination in the **Global LFO** section is at 0 at launch, so the LFO reaches nothing until one of them comes up.*

The section

Top to bottom: **LFO Rate** and **LFO Rate Range**, **LFO Waveform**, then a **Second Oscillator** group holding **LFO 2 Depth**, **LFO 2 Rate Offset (vs. LFO 1)**, **LFO 2 Waveform**, **LFO 2 Phase Offset**, and **LFO Combine**. Below those, **Reset LFO Phase**, then **Bus Spread**, then ten destination sliders.



The Global LFO section. One signal built at the top, ten destinations below it.

11-01 The Global LFO section. One signal built at the top, ten destinations below it.

Everything above **Bus Spread** builds one signal. Everything below it decides where that signal goes.

LFO Rate

How fast the modulation runs, -1 to 1, default 0.274, with a squared response.

In **Slow** the default is one full cycle in about 42 seconds, and full travel is a cycle in about 3 seconds. At 0 the LFO stops and holds, and every destination sits at the value it was holding when it stopped. Below 0 it runs backwards.

One cycle in four seconds is 0.886 in **Slow** and 0.280 in **Fast**.

LFO Rate Range

Two entries, **Slow** and **Fast**, default **Slow**. **Fast** multiplies the whole slider by ten: the default becomes a cycle in about 4 seconds and full travel is about three cycles a second. It scales **LFO 2 Rate Offset (vs. LFO 1)** by the same amount, so the beat between the two oscillators keeps its proportion to the carrier.

LFO Waveform

The five waveforms from chapter 5, under **Sine**, **Tri**, **Saw**, **Sqr**, and **S&H**, default **Sine**.

Sine and Tri travel smoothly out and back. Saw runs one way and resets, so a modulated control sweeps and then jumps home. Sqr holds one extreme, then the other, with nothing in between, which turns any destination into a two-state switch. S&H steps to a new value and holds it, about six times for every cycle the other four complete.

Selecting **S&H** reveals **LFO S&H Slew / Lag Time**, 0 to 1, default 0. At 0 each step is instant. Raise it and each step becomes a ramp into the next value, until at 1 the ramp fills the whole step and the modulation never rests.

Second Oscillator

A second oscillator, combined with the first into the one signal every destination reads.

LFO 2 Depth sets how much of it goes into that combine, 0 to 1, default 0. At 0 in **Add** or **Sub**, LFO 1 is the whole signal.

LFO 2 Rate Offset (vs. LFO 1) is a frequency difference, -1 to 1, default 0. It is added to whatever **LFO Rate** is set to, so the gap between the two oscillators stays put when the carrier is swept. At either extreme in **Slow** the two beat against each other about every 13 seconds; near zero the beat stretches out into minutes. **Fast** shortens all of that by a factor of ten.

LFO 2 Waveform offers the same five entries and its own **LFO 2 S&H Slew / Lag Time**, on the same terms as LFO 1's.

LFO 2 Phase Offset slides LFO 2's wave against LFO 1's, 0 to 1, default 0, where 1 is a full cycle. At 0.5 the two are opposite. Neither rate changes, so this shifts the alignment without disturbing a beat that is already running.

LFO Combine

How the two oscillators become one signal. Four entries: **Add**, **Sub**, **XOR**, **Mult**, default **Add**.

Add sums them, which is what beats. **Sub** subtracts LFO 2, which beats the other way up.

XOR takes the distance between the two waves. The result is always positive, so modulation that used to swing either side of the hand setting now only ever pushes one way, and it folds at every crossing, which reads as

roughly double the rate. It is the one entry that changes the signal at **LFO 2 Depth 0**, where it leaves the size of LFO 1 alone rather than LFO 1 itself.

Mult ring modulates the two together: motion clusters where both waves are strong and falls away between them. At **LFO 2 Depth 0** it silences the modulation entirely and no destination moves.

The signal is held between -1 and 1. In **Add**, with both oscillators up, the peaks flatten off against that limit while the middle of the travel keeps its shape.

Reset LFO Phase

Re-aligns both oscillators to the same starting point, so the beat between them begins again from coincidence. It also re-aligns the sweep behind **LFO → Capture Phase**.

With **LFO Rate** at 0 this parks the signal at a repeatable value, which is what makes two captures with the LFO in them comparable.

Bus Spread

The LFO output as it reaches a destination is a bus, and each destination drives a family: a set of controls it moves together. **Bus Spread** is the phase angle between successive members of every family, -1 to 1, default 0.

At 0 all members read the same value and move in step. At 0.5 each member sits a quarter cycle from the one before it, which is what turns a pulse along a diagonal into a circle. At 1.0 successive members are half a cycle apart and counter-move: one member is at its peak while the next is at its trough. The negative half is the same angles the other way round, which reverses the direction a center path travels.

 Bus Spread at 0, 0.5, and 1.0. Two fold centers, three arrangements.

11-02 Bus Spread at 0, 0.5, and 1.0. Two fold centers, three arrangements.

Every family has an order, listed with the destinations below. In the three-member family the third member sits at twice the second's angle, so the two gaps never move together.

The destinations

Ten sliders, all -1 to 1, all default 0. Negative inverts the modulation for that destination alone. Each one sums onto the hand setting of the controls it drives, so the slider you set by hand stays the middle of the travel.

Amplitude families

LFO → Rotation Amp drives both rotations' **VCO Amplitude**. Rotation 1 leads and Rotation 2 sits one **Bus Spread** step out. At 1.00 each swings 1.5 either side of its setting.

LFO → Mirror Amp drives both mirrors' **VCO Amplitude**, Mirror 1 leading and Mirror 2 one step out, 0.3 either side of each setting. That response is cubed, so the same 0.3 reads as far more movement with the hand amplitude already high than it does near 0. The sum is not held to the slider's own range: with the amplitude near 1, the modulation pushes the fold past anything the slider reaches on its own.

 **LFO → Mirror Amp**. The ripple opens and closes once every four seconds.

11-03 **LFO → Mirror Amp**. The ripple opens and closes once every four seconds.

LFO → Displace Amp drives the X and Y **VCO Amplitude**, X leading and Y one step out, 0.25 either side, which is the whole travel of both axes. At **Bus Spread** 0.5 the two axes reach quadrature and the field moves in circles instead of pulsing on a diagonal.

 **Bus Spread 0.5 on LFO → Displace Amp**. The two axes are a quarter cycle apart and the picture travels in a circle.

11-04 **Bus Spread 0.5 on LFO → Displace Amp**. The two axes are a quarter cycle apart and the picture travels in a circle.

Center families

These drive a center as an XY path. The signal supplies X, and the same signal a quarter cycle later supplies Y, so the center travels rather than sliding along a line.

LFO → Rotation Center moves both twist origins, 0.5 either side on each axis. Rotation 1 takes the path as it stands. Rotation 2 takes it with the axes swapped, which reflects its path across the diagonal, and **Bus Spread** turns that second path against the first. Both sum onto the hand **Center X** and **Center Y**, and onto **Dynamic Orbit Depth** where it is up. See chapter 7.

LFO → Mirror Center moves both fold centers on the same terms, 0.5 either side, Mirror 2's path swapped and turned. The two seams pull apart and converge rather than crossing the frame together. See chapter 6.

LFO → Displace Center moves the one displacement center, 0.5 either side. There is no second member, so **Bus Spread** has nothing to angle this one against: X drives X and Y drives Y at any setting. See chapter 8.

Keying

LFO → Keying drives all three keyers, 0.4 either side of each. **Keyer 1 — Mirrors** leads and carries the group; **Keyer 2 — Rotation** sits one **Bus Spread** step out and **Keyer 3 — Displacement** two, so the gaps between the three thresholds breathe instead of holding.

A keyer parked at or above 0.99 cuts nothing, and this destination moves thresholds through that point, so a keyer that is silent by hand comes and goes under modulation. See chapter 10.

 **LFO → Keying**. One threshold sweep, three keyers, the gaps between them breathing.

11-05 **LFO → Keying**. One threshold sweep, three keyers, the gaps between them breathing.

Standalone destinations

LFO → Radial Push sweeps **Radial Push Direction** 0.5 either side, held inside 0 to 1, so displacement drifts between cartesian shimmer and radial breathing.

LFO → Luma Staging drifts **Luma Source Staging** 0.5 either side, held inside 0 to 1. How much of it shows depends on how much luma modulation the patch has dialed in elsewhere. See chapter 4.

LFO → Capture Phase moves where in each stop-motion trigger cycle the frame is taken. It shows only while recording in stop-motion with **LFO Auto-Trigger** armed, and it changes the spacing between captures rather than the picture. Its sweep runs sixteen times faster than the signal every other destination reads, so the capture spacing can shuffle quickly while the image drifts slowly, and it wraps rather than stopping at either end. See chapter 13.

Starting points

One breathing fold. Press **M1 only**, set Mirror 1's **VCO Amplitude** to 0.6 and **Phase Speed** to 0, then take **LFO → Mirror Amp** to 1.00 and **LFO Rate** to 0.886. The ripple opens and closes every four seconds. Switch **LFO Rate Range** to **Fast** and the same slider position is a flutter instead.

Circles from two axes. Press **Disp only**, set both **Phase Speed** controls to 0, and take **LFO → Displace Amp** to 1.00. The picture pulses along one diagonal. Now bring **Bus Spread** to 0.5 and the same modulation travels around instead, because the two axes are a quarter cycle apart. Take **Bus Spread** to -0.5 and it travels the other way.

Two oscillators beating. From the first patch, set **LFO 2 Depth** to 0.7 and **LFO 2 Rate Offset (vs. LFO 1)** to 0.2 with **LFO Combine** on **Add**. The four second breathe now grows and shrinks over a much longer cycle. Press **Reset LFO Phase** to start that cycle from the top. Switch **LFO Combine** to **Mult** and the motion goes quiet except where both waves are strong.

12. Mix and Vignette

Two stages sit after everything else. One blends the finished picture back against the source, and one crops what comes out of that.

*Before you start: **Reset All**, press **M1 only**, set Mirror 1's **VCO Amplitude** to 0.4 and its **Phase Speed** to 0, and turn **Enabled** on in **Keying** so there are holes in the frame to see through.*

Mix

Dry / Effected

Blends the finished picture against the source, 0 to 1, default 1.

At 1 you see the effected picture alone. Pull it down and the source comes up underneath everything the chain, the holes, and the keys have done, until at 0 the frame is the source and no geometry is visible at all.



Dry / Effected at 1 and at 0.35. The holes fill with the source rather than reopening.

12-01 Dry / Effected at 1 and at 0.35. The holes fill with the source rather than reopening.

The dry side is not the raw input. It is the source read through **Canvas: Zoom, Border Behavior**, and **Edge Pre-Crop (Raw Input)** all apply to it, and nothing else does. See chapter 3.

Because this is a blend rather than a switch, the parts of the frame that were cut to black do not come back as themselves. They fill with the dry picture at whatever strength the slider is set to, so at 0.4 a hole reads as a faded copy of the source sitting in the gap. See chapter 9.

Vignette

A shape matte over the mixed result, and the last thing that happens to the picture.

It is measured on the frame, not on the image inside it. **Zoom**, the chain, and the centers of every module move the picture around underneath a shape that stays where it is put.

The Vignette section with a shape selected. Five controls and the alpha toggle appear; at Off none of them are on screen.

12-02 The Vignette section with a shape selected. Five controls and the alpha toggle appear; at Off none of them are on screen.

Shape

Off, **Circle**, or **Rect**, default **Off**. The five controls below it appear when a shape is selected and are hidden at **Off**.

Outside the shape goes black.

Center X and Center Y

Where the shape sits, -0.5 to 0.5 on each axis, default 0. At 0 the shape is centered on the frame, and 0.5 puts its center on an edge.

Size

How big the shape is, 0.05 to 1.0, default 0.5, measured as a fraction of the frame rather than in pixels.

Circle at 0.5 touches the middle of all four edges and takes the corners, so on a 16:9 output it reads as an oval wider than it is tall. Past about 0.71 the circle is larger than the frame and nothing is cut. At 0.05 the picture is a small disc in the middle of black.

Rect at 0.5 is exactly the frame, so nothing is cut there. Bring it down and a black border closes in evenly on all four sides.

Circle at Size 0.35, 0.5, and 0.71.

12-03 Circle at Size 0.35, 0.5, and 0.71.

Aspect Stretch (Wide / Tall)

Stretches the shape on one axis and squeezes it on the other, -1 to 1, default 0. Negative is wide, positive is tall.

Because the shape is measured on the frame, a **Circle** on a 16:9 output is already an oval at 0. About 0.384 is where it reads as round on screen. Past that it goes tall, and at 1 it is a narrow vertical slot.

On **Rect** this same control sets the width-to-height ratio, which is why there are no separate width and height sliders.

 Aspect Stretch (Wide / Tall) at 0 and at 0.384. On a 16:9 output the second one is round.

12-04 Aspect Stretch (Wide / Tall) at 0 and at 0.384. On a 16:9 output the second one is round.

Edge Softness

Feathers the edge of the shape, 0 to 0.5, default 0.

At 0 the edge is a hard line. Above 0 the fade is centered on that line, half of it inside the shape and half outside, so the picture is at half strength exactly where the hard edge used to be. The band widens with the number, and at 0.5 on a shape of default size the fade covers the whole frame and the picture falls off continuously from the center out.

 Edge Softness at 0, 0.15, and 0.5.

12-05 Edge Softness at 0, 0.15, and 0.5.

Matte to Alpha

Writes the shape into the alpha channel as well, off by default, and available only with a shape selected.

Nothing about the picture on screen changes when this is on. The window stays opaque whatever the matte is doing, and what the control changes is what leaves the instrument: a **ProRes 4444** take carries the matte as a real alpha channel, and a PNG screenshot keeps it too. Every other codec discards it and gives you the same black outside the shape as before. See chapter 13.

The color is already multiplied down by the matte before the alpha is written, which is what compositing software expects from a premultiplied file.

Starting points

Fill the holes. From the chapter patch, bring **Collision Threshold** down to about 0.5 so the holes are large, then pull **Dry / Effected** to about 0.35. The cut regions fill with a soft copy of the source and the folded picture floats over it. Sweep the slider between 0.2 and 0.6 while the fold moves.

A round window. Set **Shape** to **Circle**, **Size** to 0.35, and **Aspect Stretch (Wide / Tall)** to 0.384 for a round frame on a 16:9 output. Bring **Edge Softness** to 0.12, then move **Center X** slowly. The picture underneath does not move with the shape, so the window travels across the image.

Two shapes at once. Same vignette, then take **Dry / Effected** to 0.5. The dry picture and the effected picture are both inside the circle, because the matte is applied after the blend and crops the two together.

13. Recording and capture

The **Recording** section is the last one in the sidebar. It holds what size the instrument runs at, what a take is written as, the transport, and the screenshot button.

*Before you start: press **Choose Output Folder...** at the bottom of the section and pick somewhere to write. Nothing records until you have.*

What gets recorded

The recorder takes the picture, not the window. The sidebar is not in the file, and neither is the cursor, the title bar, anything that passes in front of the window, or the black bars around a letterboxed preview. Screenshots come from the same place.

The vignette is inside the picture rather than over it, so it is in the file. See chapter 12.

Output Resolution

What size the instrument renders at, from a menu of eight: **Native**, **Native ÷ 2**, **640 × 480**, **1280 × 720**, **1920 × 1080**, **1080 × 1080**, **1080 × 1920**, and **3840 × 2160**. Default is **Native**.

This one setting governs the preview, recordings, and screenshots together. There is no separate recording size. The line under the picker reads back what is being produced right now.

Picking one of the six fixed sizes reshapes the window onto that aspect, as large as the screen allows. The preview is not pixel-for-pixel with the file at that point, and the file is the fixed size regardless of how big the window ends up. **Native** follows the window, and **Native ÷ 2** follows it at half the width and half the height.

Whatever the source's own shape is, it is stretched to fill this frame. A 4:3 camera in a 16:9 output comes out wide. See chapter 3.

The picker is locked while a take is running. With **Native** or **Native ÷ 2** selected, resizing the window during a take ends it: the file is saved and the status line reads that the recording stopped because the window resized.

Codec

What the next take is written with. **H.264**, **ProRes 422 HQ**, or **ProRes 4444**, default **H.264**. Locked while a take is running, and read once when **Record** is pressed.

H.264 produces small files that play anywhere.

The two ProRes entries are intra-only masters and carry Rec.709 color tagging. A ProRes take can look slightly different in some players from an H.264 take of the same material, and it is the tagged file that is being described correctly. They are also much heavier on disk, around 24 MB/s for 422 HQ at 1920 × 1080 and 30 fps, and around 150 MB/s for 4444 at 3840 × 2160 and 30 fps. A slow external or network volume at those rates backs the writer up and drops frames.

ProRes 4444 is the only entry that carries an alpha channel. The line under the picker states which of the two cases you are in, and the matte itself comes from **Matte to Alpha** in **Vignette**. See chapter 12.

Mode

Long-form or **Stop-motion**, default **Long-form**. Locked while a take is running.

Long-form

Continuous capture at the frame rate below. The transport is **Record**, then **Pause** and **Stop**, with **Pause** becoming **Resume** once pressed.

The motion in the file is spaced by the frames that were captured, so a hitch in the live view is not written into the take.

Stop-motion

Single frames accumulated into one clip. The transport is **Record** and **Stop**, with no pause, and a **Capture Frame** button that adds exactly one frame each time it is pressed. It is available only while a take is running.

The instrument keeps running between captures, and the captures are not spaced evenly in real time. That gap is what separates a stop-motion take from a long-form one at the same frame rate.

Frame Rate

Eight fixed rates: 5, 7, 10, 12, 14, 24, 30, and 60, default 30. Locked while a take is running.

In **Long-form** this is how many frames a second are captured. In **Stop-motion** it is the playback rate of the assembled clip: 60 captured frames at 12 is a five second file, and the same 60 frames at 30 is a two second file, however long you stood there taking them.

The auto-trigger

Three controls under the stop-motion transport, plus a meter. Unlike the pickers above, these stay live during a take.

LFO Auto-Trigger arms an internal clock that takes frames on its own, off by default.

Trigger Rate sets how fast that clock runs, -1 to 1, default 0.3, with a squared response. The default is about one capture every 1.2 seconds. Full travel is about nine and a half captures a second. Below 0 the clock runs backwards and still takes one frame per cycle.

Capture Phase sets where in each cycle the frame is taken, 0 to 1, default 0.5. Moving it re-times the next capture, so dragging it by hand pulls the spacing around while the clock keeps its rate. **LFO → Capture Phase** in the **Global LFO** section sweeps it for you, and that sweep wraps around rather than stopping at either end. See chapter 11.

Trigger Scope appears while the toggle is on. The dot is the clock's playhead and the line is the capture point. A frame is taken as the dot passes the line.



The stop-motion transport with the auto-trigger armed. The dot on the Trigger Scope is the clock; the line is where the frame is taken.

13-01 The stop-motion transport with the auto-trigger armed. The dot on the Trigger Scope is the clock; the line is where the frame is taken.

While a take runs

A red dot sits below the transport, solid while recording and dimmed while paused.

Long-form shows elapsed time and the number of frames written. **Stop-motion** shows the number of frames and the length of the clip they add up to at the current frame rate.

A dropped-frame count appears in orange as soon as there is one.



A stop-motion take in progress. Frames captured, and the clip they add up to.

13-02 A stop-motion take in progress. Frames captured, and the clip they add up to.

When a take ends

The status line at the bottom of the section names the file that was saved.

For a long-form take it also carries a pair of numbers: the clip's duration and the real time the take ran. On a take that kept up, these sit within a few tenths of each other. A clip meaningfully shorter than real time means the frames asked for were not delivered. Each **Pause** rounds to the nearest second, so the pair is exact only on a take that ran straight through. Stop-motion carries no such pair, since its clip length comes from the frame count and not from the clock.

Dropped frames are reported with their cause: writer means the encoder or the disk fell behind, pool means the capture side did. Both counts appear in the same line.

Quitting the app during a take finalizes it first. The file is saved and says so.

Screenshots

Screenshot writes the current frame as a PNG at **Output Resolution**, whether or not a take is running. The status line names the file and its dimensions.

A screenshot keeps the alpha channel when **Matte to Alpha** is on, which no codec but ProRes 4444 does.

Where files go

Choose Output Folder... picks the destination, and the button then reads that folder's name. Both recordings and screenshots go there.

Record refuses to start without a folder and says so. **Screenshot** opens the chooser instead.

Files are named for the date and time to the second, ending in `.mov` for a take and `.png` for a screenshot, with a number added if that name already exists.

macOS sometimes blocks writing to folders inside Desktop, Documents, and Downloads even after you have picked one. That is tested before a take starts rather than discovered at the end, and the status line names the folder and suggests another.

Starting points

A clean four second loop. **Output Resolution** 1920 × 1080, **Codec** ProRes 422 HQ, **Mode** Long-form, **Frame Rate** 60. Set **LFO Rate** to 0.886 with any destination up, press **Reset LFO Phase**, then **Record**, and **Stop** after about ten seconds. The take contains several whole cycles to cut a four second loop from, and the status line's two numbers say whether the machine kept up.

Stop-motion by hand. **Mode** Stop-motion, **Frame Rate** 12. Press **Record**, then move one control and press **Capture Frame**, and repeat. Twelve presses make one second of film. The frame counter and the clip length beside it tell you how far in you are.

Let the clock take them. Same patch, turn **LFO Auto-Trigger** on and set **Trigger Rate** to about 0.5. Frames start arriving on their own. Now take **LFO → Capture Phase** in the **Global LFO** section to 1.00 with **LFO Rate** low: the captures stop being evenly spaced, and the picture jumps further between some frames than others.

14. MIDI

A hardware controller can drive nearly every control in the instrument. The bindings are made in a separate window and are kept between sessions.

*Before you start: connect a controller, then open the mapping window with the **MIDI...** button at the top of the sidebar, the **MIDI Mapping...** menu item, or ⌘M.*

The mapping window

The window is closed at launch and can stay closed while you play. The MIDI engine runs whether or not it has ever been opened, so a mapping made once keeps working with the window shut.

Its layout, top to bottom: a list of sections on the left with that section's parameters on the right, a detail strip under the parameter list once a row is selected, then **MIDI Devices**, then **MIDI Glide**, then a status line. **Clear All Mappings...** is in the toolbar.



The mapping window. Sections on the left, parameters on the right, and the device list, glide, and status line along the bottom.

14-01 The mapping window. Sections on the left, parameters on the right, and the device list, glide, and status line along the bottom.

Twelve sections are listed, in sidebar order from **Canvas** down to **Recording**. **Source** is not among them: which source is playing is not a mappable parameter.

Nothing in this window is typed. Bindings are made by listening, and every value is set with a slider or a menu.

Learn

Press **Learn** on a parameter's row. The button reads **Listening...**, and the next CC or Note On from any enabled device becomes that parameter's binding. Press it again to stop listening without binding anything.

The row then shows what it is bound to, as a channel and a number: Ch 3 CC 21 . An unbound row shows a dash.

Learn records the channel it saw rather than binding to all of them. Widening it to any channel is done in the detail strip below.

One CC drives one parameter. Learning a CC that is already bound elsewhere takes it from the old parameter, and that row goes back to a dash.

A new binding starts with the parameter's full range under the fader.

The detail strip

Select a row and the strip appears underneath the list. It carries the parameter's name, a **Channel** picker offering **Any** or **Channel 1** through **Channel 16**, and the range controls.

Low and **High** set the two ends of the fader's travel, in the parameter's own units. Narrowing them puts the whole fader across the part of the range you want. Setting **Low** above **High** inverts the control, and the strip says so when you do.

A fader spreads its travel evenly across the numbers between **Low** and **High**, which is not the same as the travel of the slider on screen: several sliders in the instrument are curved. On those, the fader's middle does not land where the on-screen slider's middle does, and **Low** and **High** are how you get the span you want back.

Toggles have no range to set. A CC of 64 or above turns one on and anything below turns it off, and a Note On flips it from wherever it is.

Pickers take **Low** and **High** as entries rather than numbers, and the readout names the entry rather than showing a figure. A fader on **LFO Waveform** with **Low** at Sine and **High** at Tri reaches those two and nothing else.

The detail strip. Channel, then Low and High in the parameter's own units.

14-02 The detail strip. Channel, then Low and High in the parameter's own units.

Two fixed notes

Two actions are on notes rather than in the parameter list, on any channel, with nothing to learn:

- **Note 0** takes a stop-motion frame, the same as pressing **Capture Frame**.
- **Note 1** resets the LFO phase, the same as pressing **Reset LFO Phase**.

See chapter 13 and chapter 11.

MIDI Devices

Every MIDI source the system offers is listed by name with a checkbox. Unchecking one stops it reaching the instrument; checking it again resumes, with its mappings intact.

A controller connected after launch appears in the list on its own and picks up the mappings it already had, with no relaunch. If the list is empty, the window says so and points at Audio MIDI Setup.

The instrument also publishes a destination named **Mirrororor**, so software that sends MIDI can address it directly.

What can be mapped

Every parameter in the twelve sections except nine, which are listed in place, greyed, with the reason on the row.

Output Resolution is out because each value rebuilds the frame the instrument renders into. **Frame Rate**, **Codec**, and **Mode** are read once when **Record** is pressed and locked for the take. The five **Geometry Order Routing** slots are out because sweeping a fader through a chain order is not a gesture; the preset row is there for reordering. See chapter 4.

MIDI Glide

A single slider in the window's footer, 0 to 3 seconds, starting at **Off**.

A fader sends its position in 127 steps. At **Off** the parameter takes each step as it arrives. Above 0 each incoming value becomes a target the parameter travels toward, arriving the stated time after the fader stops moving, so a quick sweep lands as one continuous move.

It smooths incoming MIDI and nothing else. The on-screen sliders, the LFO, and every oscillator are untouched by it, and toggles and pickers are unaffected. The useful settling range is a few hundred milliseconds, which is why the slider's lower travel is spread out.

Dropping it back to **Off** lands everything in flight on its target immediately.

The setting is kept with the mappings.

The status line

The line along the bottom carries how many sources are connected, a count of packets received, and the last message the instrument acted on. A fault appears in orange at the right.

The packet count answers one question that nothing else can: whether anything is arriving at all. If a controller is being played and the count does not move, the problem is between the controller and the machine rather than in the mapping.

The sidebar does not follow

When MIDI moves a parameter, the picture changes but the on-screen slider keeps showing the number it had.

Refresh Controls, 𐀀𐀁R, re-reads every control and brings the sidebar back into agreement.

Only the readout is stale. What the instrument is doing is always the value MIDI last sent.

Mappings are kept

Bindings are written to disk as they are made and reload at launch.

Reset All does not touch them: a patch reset leaves the controller mapped exactly as it was. **Clear** on a row removes one binding. **Clear All Mappings...** in the toolbar removes every one of them, behind a confirmation.

Starting points

One fader, one fold. Open the window, go to **Mirror 1**, press **Learn** on **VCO Amplitude**, and move a fader. The row fills in. Now select that row and set **Low** to 0.2 and **High** to 0.8, so the fader's whole travel sits in the part of the range that reads as movement rather than spending half its length below the threshold of visibility.

A fader that changes waveform. **Learn** on **Mirror 1**'s oscillator picker, then set **Low** to Sine and **High** to Sqr in the detail strip. The fader now steps through three waveforms across its travel and cannot reach S&H.

Take the steps out. With a fader mapped to something continuous, set **MIDI Glide** to about 0.15 s and sweep it. The parameter stops arriving in increments. Take it to 3 s and the same sweep becomes a slow settle that finishes long after your hand has stopped.

15. Working with it

Before you start: **Reset All**. Every patch here is built from defaults, and each one names the slots it fills.

Five patches

Each block below is a whole patch, control by control, and one thing to move once it is running. The chapters they draw on are linked where a control needs more than a line.

The room, folded

Live Camera. Press **Mir→Rot** in **Geometry Order Routing**, which fills slot 2 with Mirror 2, slot 3 with Mirror 1, slot 4 with Rotation 2, and slot 5 with Rotation 1. Set Mirror 1's **VCO Amplitude** to 0.35 and Mirror 2's to 0.25, and both **Phase Speed** controls to 0.15. Leave Rotation 1's **VCO Amplitude** at its default 0.5 and Rotation 2's at 0. Pull **Dry / Effected** down to 0.75.

The room arrives in quadrants, folded across two lines at right angles, with the twist smearing the outer edges. The dry picture underneath keeps the furniture recognisable, so it stays clear which part of the frame each fold came from.

The room, folded. Two mirrors and a twist over a dry camera picture.

15-01 The room, folded. Two mirrors and a twist over a dry camera picture.

Now bring Rotation 2's **VCO Amplitude** to about 1.0 and set its **Frequency Offset (vs. Rotation 1)** to -5. Rotation 2's twist goes uniform and the whole frame swings as one piece under the folds, first one way and then the other.

Frozen geometry, drifting color

Gradient Synth. Press **M1 only**, then set slot 2 to Mirror 2. Mirror 1: **VCO Amplitude** 0.5, **VCO Frequency** 3, **Phase Speed** 0. Mirror 2: **VCO Amplitude** 0.4, **VCO Frequency** 7, **Phase Speed** 0. Turn **Enabled** on in **Keying** and set **Collision Threshold** to 0.6.

Both waves stand still, so the two seams and the grid of holes between them hold exactly where they are. What moves is the color: the gradient's three fields keep sliding through the frozen shape, and the pattern fills with different color every few seconds.

 Two mirrors with both Phase Speed controls at 0. The seams and the holes hold still while the gradient's color moves through them.

15-02 Two mirrors with both Phase Speed controls at 0. The seams and the holes hold still while the gradient's color moves through them.

Nothing brings the frame completely to rest. **Phase Speed**, **Spin**, and **LFO Rate** all stop when they reach 0, and the gradient still drifts, as does any rotation in the chain, at about one cycle every thirty seconds. A still frame comes from **Capture Frame**, which is chapter 13.

Move **Mirror Angle** on Mirror 2 in steps and press **Capture Frame** at each one. The seams cross at a different angle in every shot and the hole grid re-spaces itself to match.

Liquid

Live Camera. Press **Disp only**. On the X axis set **VCO Amplitude** to 0.10, **VCO Frequency** to 24, and **Phase Speed** to 0.3. On the Y axis set **VCO Amplitude** to 0.08, **VCO Frequency** to 18, and **Phase Speed** to -0.2. Set **Oscillator Mode** to **Radial** and **Radial Push Direction** to 0.6. Bring **Luma → Displace X** to 0.15.

The picture stops sitting still on its own coordinates. The two axes crawl against each other at different rates and in opposite directions, the radial push bends that crawl into the frame's center, and the luma modulation makes everything bright in the room push harder than everything dark.

 Radial Push Direction at 0 and at 1, everything else identical.

15-03 Radial Push Direction at 0 and at 1, everything else identical.

Sweep **Radial Push Direction** from 0 to 1 by hand. At 0 the field shears along the axes, at 1 it breathes in and out of the center, and the middle is both at once. Take **LFO → Radial Push** to 1.00 with a low **LFO Rate** and that sweep runs on its own, which is chapter 11.

Rings, and holes cut by rings

Gradient Synth. Press **Rot1 only**, then set slot 2 to Rotation 2. Rotation 1: **VCO Amplitude** 1.0, **VCO Frequency** 9. Rotation 2: **VCO Amplitude** 1.0, **Frequency Offset (vs. Rotation 1)** -3. Turn **Enabled** on in **Keying**, take **Rotation → Holes** to 1.00, set **Rotation 1 + 2 Combine** to **XOR**, **Collision Threshold** to 0.55, and **Key Softness (Global)** to 0.12. In **Vignette**, set **Shape** to **Circle**, **Size** to 0.4, and **Edge Softness** to 0.1.

There is no mirror in the chain, so the rings are the whole hole signal and **Wave Folding** does nothing here. The two ring sets run at different frequencies and the XOR cuts where they disagree, which puts the holes in concentric bands that walk outward as the two rotations drift apart. The soft key turns every band edge into a fade, and the vignette crops the result to an oval on a 16:9 output. Chapter 12 has the control that rounds it.

 Two rotations into the hole cutter, combined by XOR, inside a soft vignette.

15-04 Two rotations into the hole cutter, combined by XOR, inside a soft vignette.

Step **Rotation 1 + 2 Combine** through its four modes with everything else untouched. **Additive** thickens the bands, **Subtractive** thins them, **Multiply** leaves holes only where both ring sets are strong at once, and **XOR** is where you started.

Four seconds that repeat

Gradient Synth. Press **M1 only**. Mirror 1: **VCO Amplitude** 0.6, **Phase Speed** 0, **Spin** off. In **Global LFO** set **LFO → Mirror Amp** to 1.00, **LFO Rate Range** to **Slow**, and **LFO Rate** to 0.886, which is one full cycle in four seconds. Press **Reset LFO Phase**. In **Recording** set **Output Resolution** to 1920 × 1080, **Codec** to ProRes 422 HQ, **Mode** to Long-form, and **Frame Rate** to 60. Record about ten seconds.

The only control moving the geometry is the LFO, so the fold opens and closes on a four second cycle and every four seconds the shape is identical. Cut the loop anywhere in the take. What does not repeat is the picture going in: the gradient drifts on its own clock, so the shape lands on slightly different color each pass.

Add Mirror 2 to slot 2 with **VCO Amplitude** 0.6 and **Phase Speed** 0, then take **Bus Spread** to 0.5. Mirror 2 now opens a quarter cycle behind Mirror 1 and the two seams take turns. The loop is still four seconds, because both mirrors are reading the same LFO.

 Two mirrors on one LFO at Bus Spread 0.5. The seams take turns and the cycle repeats every four seconds.

15-05 Two mirrors on one LFO at Bus Spread 0.5. The seams take turns and the cycle repeats every four seconds.

Behavior that is hard to find by accident

Where two keyers disagree

Two keyers at the same threshold, on different Feeds, with both **XOR** checkboxes on, cut only where their two feeds disagree about which side of the threshold a point sits on. Everywhere the feeds agree, cut lands on cut and cancels.

That is a difference key, and it is built in chapter 10 control by control. No single control produces it. It falls out of two keyers, two taps, and two checkboxes set together, and the black it leaves moves whenever either feed moves.

Modules that are not in the chain still make signal

An empty slot is a real bypass for the picture, and not for anything else.

Dynamic Orbit Depth on either rotation moves the twist center along a path driven by the two mirrors' oscillators. Those mirrors do not have to be in the chain, and their **Enabled** switches do not have to be on. Take a rotation's **Dynamic Orbit Depth** to 1, then go to **Mirror 1** and move its **Phase Speed** and **VCO Amplitude**: the orbit changes speed on one axis and widens as the amplitude climbs, with nothing folded anywhere in the frame. Mirror 2 does the same on the other axis. See chapter 7.

The keyers read their families the same way. A keyer set to its family Feed sees what that family alone does to the source, chain or no chain, so **Keyer 1 — Mirrors** cuts by a fold that is nowhere on screen. See chapter 10.

The dry side is a clean picture

Dry / Effected at 0 is the source with **Zoom**, **Border Behavior**, and **Edge Pre-Crop (Raw Input)** applied and no warp at all. Set **Border Behavior** to **Tile Reflect**, pull **Dry / Effected** to 0, and the frame is a clean mirrored tiling of the source with none of the chain in it. Bring the slider back up and the warped picture arrives over the top of that tiling rather than replacing it. See chapter 12.

Starting points

One fader over a finished patch. Build **The room, folded**, press **MIDI...** at the top of the column, and **Learn** a fader onto Mirror 1's **VCO Amplitude** with **Low** 0.1 and **High** 0.6. Set **MIDI Glide** to about 0.5 s. The fold now opens and closes under your hand and coasts to a stop instead of arriving in steps.

Twelve frames a second by hand. Build **Liquid**, set **Mode** to Stop-motion and **Frame Rate** to 12, and press **Record**. Move **Center X** a little and press **Capture Frame**, and repeat. Twelve presses make one second, and the camera keeps moving between them, so each frame carries a different room as well as a different push.

16. Troubleshooting

Nothing needs setting up for this chapter. It is ordered by what is in front of you rather than by section.

The frame is black

Work down this list.

Video File is selected and no file is loaded. The button under the picker reads **Load Video...** until there is one. See chapter 3.

Live Camera is selected and the status line under the picker is red. That is a fault, and the strings are in the next section.

Live Camera is selected, the line begins `Live -`, and the frame is still black. The camera has started and no picture has arrived from it yet. Give it a moment, then select **Live Camera** again.

Enabled is on in **Keying** with **Invert Key** on and nothing else cutting. Invert swaps cut for kept across the whole shape, so a frame with nothing cut becomes a frame with nothing kept. Turn **Invert Key** off, or bring **Keyer 1 — Mirrors** down until there is something to invert. See chapter 10.

Keyer 1 — Mirrors is near 0 with **Darks** off. The cut grows darker-ward as the number falls, and near 0 it has taken everything in its feed that is not already black.

Collision Threshold is near 0.05. Every peak in the hole signal is cut and the holes have met each other. See chapter 9.

Vignette is on with **Size** near 0.05, or with **Center X** or **Center Y** near an edge. The shape is small or mostly off the frame, and outside the shape is black. See chapter 12.

Pull **Dry / Effected** to 0 as a test. The source arrives with no geometry, no holes, and no keys in it, so a frame that is still black is a source problem and a frame that comes back is one of the stages above.

The camera

Chapter 3 prints the healthy sequence: **Requesting access...**, then **Starting...**, then a line beginning `Live -` with the device's name and the resolution it settled on. Anything else turns the line red and it stays there.

Selecting **Live Camera** again re-runs the whole sequence from the top: the permission check, the search for a device, the setup, and the start. That is the retry.



The camera status line in fault. Red, and it stays there until the source is selected again.

16-01 The camera status line in fault. Red, and it stays there until the source is selected again.

Access denied

```
Access denied. System Settings > Privacy & Security > Camera
```

macOS is not letting the instrument see a camera. It arrives if you refused the prompt, if the app's camera switch is off, or if the machine's policy forbids it. Turn the switch on at the path the line names, then select **Live Camera** again. macOS asks once and does not ask twice, so after a refusal the switch is the way back.

No camera found

```
No camera found
```

The search returned nothing. Connect a camera and select **Live Camera** again, which searches afresh.

Cannot open

```
Cannot open FaceTime HD Camera: ...
```

The device was found and would not open. The name before the colon is the device that was tried, and the rest of the line is the system's own description of what failed. Two of them are the instrument's own words instead:

```
Cannot open FaceTime HD Camera: input rejected by session
```

```
Cannot open FaceTime HD Camera: output rejected by session
```

The device opened and then would not attach. Select **Live Camera** again to build from scratch.

The same shape appears if a camera that was running stops, which is what unplugging one mid-picture gives you. The failed session is discarded rather than kept, so selecting **Live Camera** again builds a new one.

A control that appears to do nothing

Four questions, cheapest first.

Is the module in the chain?

A module left out of the chain keeps every one of its controls live and none of them reach the picture. **Geometry Order Routing** names them under the slots:

Not in chain: Displacement

Two things read a module the chain has left out. **Dynamic Orbit Depth** on either rotation is driven by the two mirrors' oscillators whether or not those mirrors are placed or switched on, and a keyer set to its family Feed reads that family with the chain taken out of it. See chapter 15.

Is the stage it belongs to switched on?

Enabled at the top of **Keying** gates that whole section, the hole cutter included, and it is off at launch. **Collision Threshold**, **Wave Folding**, **Rotation → Holes**, **Rotation 1 + 2 Combine**, all three keyers, **Darks**, **Key Softness (Global)**, and **Invert Key** do nothing until it is on. See chapter 10.

Each mirror has an **Enabled** switch of its own. A mirror switched off does not fold, adds nothing to the hole signal, and drops out of the **Mirrors** tap, wherever it sits in the chain. See chapter 6.

Does it have anything to act on?

Wave Folding acts on mirror interference. A chain with no mirror in it has none, so the control does nothing at any setting, whatever else is running. See chapter 9.

Rotation 1 + 2 Combine on **Multiply** outputs nothing until Rotation 2 has some amplitude.

Border Behavior and **Edge Pre-Crop (Raw Input)** do nothing on the **Gradient Synth**, which is defined everywhere in the frame and never runs out of picture. On the other two sources, **Border Behavior** shows nothing until a coordinate lands outside the picture: take **Zoom** below 1.0, or put a module in the chain. See chapter 3.

Luma Source Staging has no destination unless a luma control is up somewhere. Those are **Luma → Pivot** on the mirrors, **Luma → Twist Modulator** on the rotations, and **Luma → Displace X** and **Luma → Displace Y** on Displacement. It also does nothing with one module in the chain, and nothing when the module carrying the luma control sits in slot 5, since that one reads the source at any setting. See chapter 4.

Bus Spread sets the angle between successive members of a family, so a family with one member has nothing to angle. **LFO → Displace Center** is that family: X drives X and Y drives Y at every setting. See chapter 11.

Every **Global LFO** destination is at 0 at launch, so nothing above **Bus Spread** changes the picture until one of them comes up. **LFO Rate** at 0 stops the signal and holds it where it stopped. **LFO 2 Depth** at 0 leaves LFO 1 as the whole signal in **Add** and **Sub**, which makes **LFO 2 Waveform**, **LFO 2 Rate Offset (vs. LFO 1)**, and **LFO 2 Phase Offset** inert in those two modes.

Is it at a setting where it is silent?

A keyer cuts nothing at a working threshold of 0.99 or above, and **Keyer 1 — Mirrors** starts at 1.00.

Keyer 2 — Rotation and **Keyer 3 — Displacement** are offsets, held between 0 and 1 after they are added to **Keyer 1 — Mirrors**. Once the sum has reached either end, the rest of the slider's travel lands on the same place. That clamp is how one keyer is silenced while the others stay live, and it is the same clamp that makes the end of the slider look dead. See chapter 10.

VCO Amplitude at 0 bypasses a rotation or Displacement, and only with that module's luma controls at 0 as well. On a mirror it is not a bypass: the fold still happens with a straight fold line, and the hole signal does not read amplitude at all. Take both mirrors to 0 with **Enabled** on in **Keying** and the frame has two hard folds and a full hole pattern in it. See chapter 6 and chapter 9.

An Empty slot is the bypass that removes a module from the picture entirely. See chapter 4.

Two that change nothing on screen

Matte to Alpha in **Vignette** changes what leaves the instrument. The window is identical with it on and off, and the difference is in a **ProRes 4444** take or a PNG screenshot. See chapter 12.

MIDI Glide smooths incoming MIDI. The on-screen sliders, the LFO, and every oscillator are untouched by it, and toggles and pickers take their values whole at any setting. See chapter 14.

Recording

The take will not start

Pick an output folder first.

Record was pressed with no destination. Press **Choose Output Folder...** at the bottom of the section. **Screenshot** opens the chooser for you instead of refusing.

Can't write to this folder

macOS refused a test write to the folder you picked. The rest of the line names the folder and says that folders inside Desktop, Documents, and Downloads are sometimes blocked even after they have been chosen. Pick a different one. The test runs before the take rather than at the end of it.

```
Not ready yet – try again in a moment.
```

Record was pressed before a frame had been drawn at the current size. Press it again.

```
Could not start the recording file.
```

The file could not be opened for writing. Check there is room on the volume, then pick the folder again.

The clip is not the length you expected

In **Stop-motion**, clip length is the number of frames divided by **Frame Rate** and has nothing to do with how long you stood there. Sixty captures at 12 is a five second file whether they took a minute or an hour. See chapter 13.

In **Long-form** the status line carries both numbers when the take ends:

```
Saved 2026-08-12_14-12-03.mov (clip 9.8s / real 10.0s)
```

The first is what is in the file and the second is how long the take ran. Within a few tenths of each other, the recorder kept up. A clip meaningfully shorter than real means the frames asked for were not delivered. Each **Pause** rounds to the nearest second, so the pair is exact only on a take that ran straight through.

A take that ended on its own says why in front of the file name:

```
Recording stopped: window resized – saved ...
```

```
Recording stopped: app quit – saved ...
```

The first happens with **Output Resolution** on **Native** or **Native ÷ 2**, where the window's size is the size the instrument renders at. Pick one of the six fixed sizes for a take that should survive a drag. The second is a take being finished at quit rather than lost.

Frames were dropped

A dropped-frame count appears in orange while the take runs, and again at the end split by cause:

```
Saved 2026-08-12_14-12-03.mov (clip 9.6s / real 10.0s) (14 dropped frames – 12 writer, 2 pool)
```

The status line at the end of a long-form take, with the clip and real pair and the dropped-frame split.

16-02 The status line at the end of a long-form take, with the clip and real pair and the dropped-frame split.

Writer means the encoder or the disk fell behind. Pool means the capture side did. Both respond to the same four changes: lower **Frame Rate**, lower **Output Resolution**, take **Codec** to **H.264**, or write somewhere faster. ProRes 422 HQ runs around 24 MB/s at 1920 × 1080 and 30 fps and ProRes 4444 around 150 MB/s at 3840 × 2160, which a network volume will not hold. See chapter 13.

A screenshot did not appear

Five lines come from writing the file, and all of them mean the frame was taken and not saved:

```
No output folder selected.
```

```
Could not read the captured frame.
```

```
Could not build the image.
```

```
Could not create 2026-08-12_14-12-03.png.
```

```
Failed to write 2026-08-12_14-12-03.png.
```

Check the folder is still there and the volume still mounted, then pick it again with **Choose Output Folder....**

The controller does nothing

The mapping window's status line carries a count of packets received. Play the controller and watch that count. If it does not move, nothing is arriving and the mapping is not what is wrong. If it moves and the picture does not, the mapping is.

With packets arriving, work through the window:

- The device is listed under **MIDI Devices** with its checkbox on. An unchecked device is cut off from the instrument and keeps its mappings.

- The row shows a binding rather than a dash. Learning a CC that is already bound elsewhere takes it from the old parameter, so a mapping that stopped working may have been taken by a later **Learn**.
- The binding's **Channel** matches what the controller sends. **Learn** records the channel it saw. Widen it to **Any** in the detail strip.
- **Low** and **High** are not sitting on a span too narrow to see.

With nothing arriving and no device listed:

No MIDI sources found. Check the device is connected and appears in Audio MIDI Setup.

A controller connected after launch appears on its own with no relaunch, so a list that stays empty is upstream of the instrument. Faults appear in orange at the right of the same status line, each carrying the system's own status number. See chapter 14.

Performance

The frame rate is fixed at 60 and there is no control for it.

What the instrument renders at is **Output Resolution**, not the size of the window. On the six fixed sizes, dragging the window bigger changes what you are looking at and not what is being made. **Native** and **Native ÷ 2** follow the window, so on those two the window is the setting. 3840 × 2160 is four times the pixels of 1920 × 1080.

A recording is where a machine that is not keeping up says so, in the clip and real pair and the drop count. Both are above, under Frames were dropped.

Things that look like faults

Nothing in the frame comes completely to rest. **Phase Speed**, **Spin**, and **LFO Rate** all stop at 0. The gradient drifts on its own and so does any rotation in the chain, at about one cycle every thirty seconds, with no control over either. A still frame comes from **Capture Frame**. See chapter 15.

A slider shows one number and the picture shows another. MIDI moves the parameter without moving the control on screen. **Refresh Controls**, ⌘⇧R, re-reads every control and brings the column back into agreement. What the instrument is doing is always the value MIDI last sent.

The patch is gone after a relaunch. No patch is written to disk. The MIDI mappings and the output folder are kept, and every control comes back at its default.

Reset All left the controller mapped. It does not touch the mappings. **Clear All Mappings...** in the mapping window's toolbar removes them, behind a confirmation. See chapter 14.

Switching source changed nothing else. The chain, the keying, the LFO, and every slider hold their values, and the new picture arrives into the patch you already had. See chapter 3.

The recording does not match the window. The recorder takes the picture rather than the window, so the sidebar, the cursor, the title bar, and the black bars around a letterboxed preview are not in the file. A hitch in the live view is not written into the take either, because the motion in the file is spaced by the frames that were captured. See chapter 13.

Starting points

Get back to a picture you can read. **Reset All**, press **M1 only** in **Geometry Order Routing**, and set Mirror 1's **VCO Amplitude** to about 0.4. One module, one fold, **Keying** off, no vignette. Now put back what you were doing one control at a time. The control that takes the picture away is the one you were on.

Prove the recorder end to end. Choose **Output Folder...**, then **Output Resolution** 1920 × 1080, **Codec** H.264, **Mode** Long-form, **Frame Rate** 30. Press **Record**, wait ten seconds, press **Stop**. The status line names the file and prints clip and real within a few tenths of each other. If it does not, the machine did not hold those settings, and **Output Resolution** is the first thing to bring down.

17. Glossary

Nothing needs setting up for this chapter.

Every term here is defined properly in one chapter and used freely everywhere after it. This says what each one means and points at that chapter. The index says where else it appears.

Control names are not in this list. Those are in Appendix A, with their ranges and defaults.

bus

The Global LFO's finished signal, as it reaches a destination. The two oscillators and **LFO Combine** build one signal between them, and every destination slider reads that same signal rather than one of its own. See chapter 11.

capture phase

Where in each trigger cycle a stop-motion frame is taken. The trigger clock runs on its own at **Trigger Rate**, and **Capture Phase** sets the point along that cycle the grab lands on. **LFO → Capture Phase** sweeps that point while the clock keeps its rate. See chapter 13.

chain

The five ordered positions the geometry modules run in. Slot 1 acts on the source and slot 5 is seen last. Everything after the chain reads what the chain produced. See chapter 4.

composite

The picture as the chain has warped it so far. A module in slot 3 acts on the composite that slots 1 and 2 made, not on the source. **Luma Source Staging** is what decides whether a module's luma control reads the composite or the source. See chapter 4.

Displacement

One of the three module families, and the only one holding a single module. It pushes the picture and preserves nothing about where a pixel started. **Keyer 3 — Displacement** taps it, and **LFO → Displace Amp**, **LFO → Displace Center**, and **LFO → Radial Push** drive it. See chapter 8.

family

A set of controls treated as one group. Each Global LFO destination drives a family, and **Bus Spread** sets the phase angle between its members. The three keyers are fixed to the three module families, one each. See chapter 11.

fold

One mirror operation across one line. Everything on one side of that line is replaced with a copy of the other side. See chapter 6.

fold line

The line a fold reflects across. It passes through the mirror's **Center X** and **Center Y**, lies at **Mirror Angle**, and takes its shape from the module's oscillator. See chapter 6.

key

A threshold decision that cuts part of the picture out and leaves it black. Everywhere the signal a keyer reads sits on the cutting side of its threshold, the picture goes. **Darks** sets which side that is. See chapter 10.

Mirrors

One of the three module families, holding Mirror 1 and Mirror 2. Their waves are also the interference the hole cutter reads. **Keyer 1 — Mirrors** taps them, and **LFO → Mirror Amp** and **LFO → Mirror Center** drive them. See chapter 6.

push

Coordinate displacement with nothing preserved. A pixel moves by an amount, in a direction, and lands wherever that puts it, so it keeps neither its position nor its distance from any center. That is what separates it from a twist. See chapter 8.

Rotation

One of the three module families, holding Rotation 1 and Rotation 2. **Keyer 2 — Rotation** taps them, and **LFO → Rotation Amp** and **LFO → Rotation Center** drive them. See chapter 7.

seam

The visible join a fold leaves, where the two halves meet. Its shape is the module's waveform read down the fold line, so a still frame shows you the wave. See chapter 6.

slot

One position in the chain, numbered 1 to 5, holding one geometry module or Empty. A module can sit in more than one slot and runs once in each. An Empty slot passes the picture through untouched. See chapter 4.

source

The one picture that goes in, before any geometry module runs. **Gradient Synth**, **Live Camera**, or **Video File**, shaped on the way in by **Zoom**, **Border Behavior**, and **Edge Pre-Crop (Raw Input)**. See chapter 3.

spread

The phase angle between successive members of a family. **Bus Spread** sets it for every family at once: at 0 the members move in step, at 0.5 each sits a quarter cycle from the one before it, and at 1.0 they counter-move. See chapter 11.

tap

The signal a keyer reads, chosen by its Feed picker. A family tap is what that family alone does to the source, and the chain has no say in it. **Warped Final** taps the finished warp instead. See chapter 10.

twist

Angle displacement that preserves radius. Every pixel turns about a center along its own circle, and none of them change how far out they sit. See chapter 7.

VCO block

The group of oscillator controls a geometry module carries at the top of its section: **Oscillator Type**, **Oscillator Mode**, **VCO Amplitude**, **VCO Frequency**, and, on some modules, a rate. It is not identical in every module. See chapter 5.

wave argument

What an oscillator is read along, set by **Oscillator Mode**. **Normal** reads the wave across the frame and **Radial** reads it as distance from the module's center, so one waveform arrives as bands or as rings. It changes what the wave is read along and not what the wave does. See chapter 5.

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Appendix A. Complete control reference

Nothing needs setting up for this appendix. It is ordered as the sidebar is, top to bottom.

Every default below is what **Reset All** restores. The source selection and anything to do with recording are left alone by it.

For a picker, the range column lists its entries. For a switch it reads Off, On.

Numbers are the stored values. The readout beside a slider shows two decimals, so a default of 0.447 reads as 0.45 on screen.

A default written as 0 (detent) means the slider catches at zero. The band either side is remapped rather than discarded, so no value is lost.

Nine controls cannot be assigned to a MIDI fader. They are marked on their rows and listed together in chapter 14.

Above the sections

Three controls sit above every section in the column, or in the menu bar.

Reset All restores every parameter below to its default and asks first. The same action is in the Edit menu as **Reset All Parameters...**, ⌘⇧R.

MIDI... opens the mapping window. In the View menu as **MIDI Mapping...**, ⌘⇧M. See chapter 14.

Refresh Controls, ⌘⇧R, is in the Edit menu only. It makes every slider re-read the value it is already holding, which is what puts the column back in step after MIDI has moved something.

Source

Covered in chapter 3.

Control	Range	Default	What it does
Source	Gradient Synth, Live Camera, Video File	Gradient Synth	Which picture goes in. Not a parameter: Reset All leaves it where it is, and it is not in the mapping window.

Load Video... appears under the picker with **Video File** selected, and reads the file's name once one is loaded.

Canvas

Covered in chapter 3.

Control	Range	Default	What it does
Zoom	0.33 to 10	1.0	How much of the frame the source fills. Above 1 magnifies, below 1 shrinks the picture toward the center. Equal travel is an equal ratio, so 0.5 and 2.0 sit the same distance either side of 1.
Border Behavior	Black Border, Bleed Lines, Tile Screen, Tile Reflect, Gradient Synth BG	Black Border	What fills the frame wherever a coordinate lands outside the source picture. Nothing on the Gradient Synth .
Edge Pre-Crop (Raw Input)	Off, On	Off	Trims the incoming picture's four edges and stretches what is left back out to fill the frame. Nothing on the Gradient Synth .
Pre-Crop Pixels	1 to 5	3	How many pixels come off each edge. Hidden until Edge Pre-Crop (Raw Input) is on.

Geometry Order Routing

Covered in chapter 4.

Control	Range	Default	What it does
Slot 1	Mirror 1, Mirror 2, Rotation 1, Rotation 2, Displacement, Empty	Empty	Which module acts on the source. Not MIDI-mappable.
Slot 2	as above	Mirror 2	Not MIDI-mappable.
Slot 3	as above	Mirror 1	Not MIDI-mappable.
Slot 4	as above	Rotation 2	Not MIDI-mappable.
Slot 5	as above	Rotation 1	Which module is seen last. Not MIDI-mappable.
Luma Source Staging	0 to 1	0	Which picture every module's luma control reads. 0 is the source, 1 is what the chain has built so far.

Eight preset buttons sit below. They set all five slots at once and touch no module parameter.

Rotation 1

Covered in chapter 7. The oscillator controls are chapter 5.

Control	Range	Default	What it does
Oscillator Type	Sine, Tri, Saw, Sqr, S&H	Sine	The waveform the module reads.
S&H Slew / Lag Time	0 to 1	0	Ramps the start of each S&H step toward the value it holds. Hidden until S&H is selected.
Oscillator Mode	Normal, Radial	Radial	What the wave is read along: down the frame, or out from the module's center.
VCO Amplitude	-3 to 3	0.5	How far the picture turns at the wave's peak, in radians. Negative twists the other way.
VCO Frequency	1 to 15	5	How many rings of twist fit into the picture. Rotation 2's frequency is derived from this one.
Luma → Twist Modulator	-0.3 to 0.3	0 (detent)	Lets the brightness of the picture add to the twist, centered on mid grey.
Dynamic Orbit Depth	0 to 1	0	Puts the center on a moving path driven by the mirrors' oscillators. Adds to Center X and Center Y rather than replacing them.
Center X	-1 to 1	0	Where the module turns about, horizontally.
Center Y	-1 to 1	0	Where the module turns about, vertically.

Rotation 2

Covered in chapter 7. Rotation 2 has no frequency and no luma control of its own.

Control	Range	Default	What it does
Oscillator Type	Sine, Tri, Saw, Sqr, S&H	Sine	The waveform the module reads.
S&H Slew / Lag Time	0 to 1	0	Hidden until S&H is selected.
Oscillator Mode	Normal, Radial	Radial	What the wave is read along.
VCO Amplitude	-3 to 3	0	How far the picture turns at the wave's peak, in radians.
Frequency Offset (vs. Rotation 1)	-15 to 15	0	Rotation 2's frequency as a distance from Rotation 1's VCO Frequency .
Dynamic Orbit Depth	0 to 1	0	Puts the center on a moving path. Rotation 2's axes follow the mirrors the other way round from Rotation 1's.
Orbit Phase Offset	0 to 1	0	Where Rotation 2 sits along its path relative to Rotation 1, 1 being a full turn. Does nothing at Dynamic Orbit Depth 0.
Orbit Lissajous Ratio	0.25 to 4	1	The shape of the path. 1 an ellipse, 2 a figure eight, 3 three-lobed. Does nothing at Dynamic Orbit Depth 0.
Center X	-1 to 1	0	Where the module turns about, horizontally.
Center Y	-1 to 1	0	Where the module turns about, vertically.

Mirror 1 and Mirror 2

Covered in chapter 6. The two sections carry the same controls, and **Mirror Angle** is the only one with a different default between them.

Control	Range	Default	What it does
Doubled	Off, On	Off	Adds a second, flat fold through the same center, running before the rippled one.
Enabled	Off, On	On	Bypasses the module wherever it sits in the chain, and takes it out of the interference signal with it.
Oscillator Type	Sine, Tri, Saw, Sqr, S&H	Sine	The waveform the seam takes.
S&H Slew / Lag Time	0 to 1	0	Hidden until S&H is selected.
Oscillator Mode	Normal, Radial	Normal	Whether the wave is read along the fold line or out from the module's center.
VCO Amplitude	-1 to 1	0	How far the seam swings. At 0 the fold is straight. The response is steep, so most of the deflection arrives late in the travel.
VCO Frequency	0.2 to 12	5	How tightly the wave packs into the picture.
Phase Speed	-1 to 1	0.447	How fast the wave scrolls along the fold line. Squared response: the default is one radian per second and full travel is five.
Luma → Pivot	-1.5 to 1.5	0 (detent)	Lets the brightness of the picture move the fold line, centered on mid grey.
Spin	Off, On	Off	Replaces Mirror Angle with Spin Speed (Bipolar) .
Spin Speed (Bipolar)	-1 to 1	0	Sweeps the fold line's angle continuously. Squared response, full travel about one turn every three seconds. Hidden until Spin is on.
Mirror Angle	0 to 2π	0 on Mirror 1, $\pi/2$ on Mirror 2	Which way the fold line lies. Hidden while Spin is on.
Doubled Angle Offset	0 to 2π	π	How far the partner fold's line is turned from the base fold's. No detent: the reset button returns to π . Hidden until Doubled is on.

Control	Range	Default	What it does
Center X	-1 to 1	0 (detent)	Where the module's anchor sits, horizontally. The fold line passes through it and turns about it.
Center Y	-1 to 1	0 (detent)	Where the module's anchor sits, vertically.

Displacement

Covered in chapter 8. Four controls govern the module, then the section splits into **X Axis** and **Y Axis**.

Control	Range	Default	What it does
Oscillator Mode	Normal, Radial	Normal	One picker for both axes. Whether each oscillator is read across the frame or out from the module's center.
Radial Push Direction	0 to 1	0	Which way the push points. At 0 X pushes horizontally and Y vertically. At 1 X pushes outward from the center and Y around it. A blend, not a switch.
Center X	-1 to 1	0	Where the module is anchored, horizontally. One pair for the whole module.
Center Y	-1 to 1	0	Where the module is anchored, vertically.

The **X Axis** and **Y Axis** blocks each carry a full copy of the following. The labels are identical in both, and the header above each block is what tells them apart.

Control	Range	Default	What it does
Oscillator Type	Sine, Tri, Saw, Sqr, S&H	Sine	The waveform this axis reads.
S&H Slew / Lag Time	0 to 1	0	Hidden until S&H is selected on this axis.
VCO Amplitude	-0.25 to 0.25	0	How far the picture is pushed at the wave's peak, as a fraction of the frame.
VCO Frequency	0.5 to 60	5	How fine the field is. The travel is curved: the bottom quarter of the track covers 0.5 to about 3.5, the top quarter about 46 to 60.
Phase Speed	-1 to 1	0	The rate. Squared response, same maximum as the mirrors'. At 0 the field holds still.
Luma → Displace X / Luma → Displace Y	-0.3 to 0.3	0 (detent)	Lets the brightness of the picture push it. Black pushes nothing and white pushes the full amount.

Keying

The hole cutter is chapter 9 and the keyers are chapter 10. **Enabled** gates all of it.

Control	Range	Default	What it does
Enabled	Off, On	Off	Gates the whole stage, the hole cutter included.
Invert Key	Off, On	Off	Swaps cut for kept across the finished shape, holes included.
Darks	Off, On	Off	Flips which side of the threshold all three keyers cut. Does not touch the hole signal.
Key Softness (Global)	0 to 0.35	0	Feathers every threshold edge in the section, Collision Threshold with them.
Collision Threshold	0.05 to 0.95	0.7	How strong the hole signal has to be before the picture is cut.
Wave Folding	0 to 2	0	Folds the mirror interference back on itself, so one hole becomes nested rings. Does nothing in a chain with no mirror in it.
Rotation → Holes	0 to 1	0	Mixes the rotations' own wave into the hole signal.
Rotation 1 + 2 Combine	Additive, Subtractive, XOR, Multiply	Additive	How the two rotations' waves become the wave above. Multiply outputs nothing until Rotation 2 has some amplitude.
Keyer 1 — Mirrors	0 to 1	1	The absolute threshold the other two are measured from. At 0.99 and above it cuts nothing.
Feed (Keyer 1)	Mirrors, Warped Final	Mirrors	Which signal this keyer reads.
XOR (Keyer 1)	Off, On	Off	Folds this keyer into the shape as XOR instead of OR.
Keyer 2 — Rotation	-1 to 1	0 (detent)	An offset from Keyer 1 — Mirrors , held between 0 and 1 after summing.
Feed (Keyer 2)	Rotation, Warped Final	Rotation	Which signal this keyer reads.
XOR (Keyer 2)	Off, On	Off	Folds this keyer in as XOR.
Keyer 3 — Displacement	-1 to 1	0 (detent)	An offset from Keyer 1 — Mirrors , held between 0 and 1 after summing.

Control	Range	Default	What it does
Feed (Keyer 3)	Displacement, Warped Final	Displacement	Which signal this keyer reads.
XOR (Keyer 3)	Off, On	Off	Folds this keyer in as XOR.

The Feed pickers carry no label on screen and each **XOR** checkbox is labelled only **XOR**; position under a keyer is what identifies them. The mapping window names the same rows Keyer 1 Feed, Keyer 1 XOR, and so on.

Global LFO

Covered in chapter 11. Everything above **Bus Spread** builds one signal and everything below it decides where that signal goes.

Control	Range	Default	What it does
LFO Rate	-1 to 1	0.274	How fast the modulation runs, with a squared response. At 0 it stops and holds. Below 0 it runs backwards.
LFO Rate Range	Slow, Fast	Slow	Fast multiplies the whole slider by ten, LFO 2 Rate Offset (vs. LFO 1) included.
LFO Waveform	Sine, Tri, Saw, Sqr, S&H	Sine	The shape LFO 1 runs.
LFO S&H Slew / Lag Time	0 to 1	0	Hidden until S&H is selected.
LFO 2 Depth	0 to 1	0	How much of LFO 2 goes into the combine.
LFO 2 Rate Offset (vs. LFO 1)	-1 to 1	0	A frequency difference added to LFO Rate , so the gap between the two holds when the rate is swept.
LFO 2 Waveform	Sine, Tri, Saw, Sqr, S&H	Sine	The shape LFO 2 runs.
LFO 2 S&H Slew / Lag Time	0 to 1	0	Hidden until S&H is selected on LFO 2.
LFO 2 Phase Offset	0 to 1	0	Slides LFO 2's wave against LFO 1's, 1 being a full cycle. Neither rate changes.
LFO Combine	Add, Sub, XOR, Mult	Add	How the two oscillators become one signal.
Bus Spread	-1 to 1	0	The phase angle between successive members of every family. 0 in step, 0.5 a quarter cycle apart, 1.0 counter-moving.
LFO → Rotation Amp	-1 to 1	0	Drives both rotations' VCO Amplitude , 1.5 either side at full. Rotation 1 leads.
LFO → Mirror Amp	-1 to 1	0	Drives both mirrors' VCO Amplitude , 0.3 either side. Mirror 1 leads.
LFO → Displace Amp	-1 to 1	0	Drives both axes' VCO Amplitude , 0.25 either side. X leads.

Control	Range	Default	What it does
LFO → Rotation Center	-1 to 1	0	Drives both twist origins as an XY path, 0.5 either side on each axis. Rotation 2 takes the axes swapped.
LFO → Mirror Center	-1 to 1	0	Drives both fold centers as an XY path, 0.5 either side. Mirror 2 takes the axes swapped.
LFO → Displace Center	-1 to 1	0	Drives the one displacement center, 0.5 either side. No second member, so Bus Spread does not reach it.
LFO → Keying	-1 to 1	0	Drives all three keyers, 0.4 either side of each. Keyer 1 leads and Keyer 3 sits two spread steps out.
LFO → Radial Push	-1 to 1	0	Sweeps Radial Push Direction 0.5 either side, held inside 0 to 1.
LFO → Luma Staging	-1 to 1	0	Drifts Luma Source Staging 0.5 either side, held inside 0 to 1.
LFO → Capture Phase	-1 to 1	0	Moves where in each stop-motion trigger cycle the frame is taken. Shows only while recording in Stop-motion with LFO Auto-Trigger armed.

Reset LFO Phase sits above **Bus Spread**. It re-aligns both oscillators to the same starting point, and the sweep behind **LFO → Capture Phase** with them.

Mix

Covered in chapter 12.

Control	Range	Default	What it does
Dry / Effected	0 to 1	1	Blends the finished picture against the source read through Canvas . At 0 no geometry is visible.

Vignette

Covered in chapter 12. Everything below **Shape** is hidden at **Off**.

Control	Range	Default	What it does
Shape	Off, Circle, Rect	Off	Which shape the matte is. Outside the shape goes black.
Center X	-0.5 to 0.5	0	Where the shape sits, horizontally. 0.5 puts its center on an edge.
Center Y	-0.5 to 0.5	0	Where the shape sits, vertically.
Size	0.05 to 1.0	0.5	How big the shape is, as a fraction of the frame. Rect at 0.5 is exactly the frame. Past about 0.71 a Circle is larger than the frame.
Aspect Stretch (Wide / Tall)	-1 to 1	0	Stretches the shape on one axis and squeezes it on the other. Negative is wide, positive is tall, and 0.384 is where a Circle reads as round on a 16:9 output. On Rect it is the width-to-height ratio.
Edge Softness	0 to 0.5	0	Feathers the edge of the shape, half the fade inside the line and half outside.
Matte to Alpha	Off, On	Off	Writes the shape into the alpha channel. Nothing changes on screen. Only a ProRes 4444 take and a PNG screenshot carry it.

Recording

Covered in chapter 13. The first four are locked while a take is running.

Control	Range	Default	What it does
Output Resolution	Native, Native ÷ 2, 640 × 480, 1280 × 720, 1920 × 1080, 1080 × 1080, 1080 × 1920, 3840 × 2160	Native	What size the instrument renders at. Governs the preview, recordings, and screenshots together. Not MIDI-mappable.
Codec	H.264, ProRes 422 HQ, ProRes 4444	H.264	What the next take is written with. Read once when Record is pressed. Not MIDI-mappable.
Mode	Long-form, Stop-motion	Long-form	Continuous capture, or single frames accumulated into one clip. Not MIDI-mappable.
Frame Rate	5, 7, 10, 12, 14, 24, 30, 60	30	Frames a second in Long-form , and the clip's playback rate in Stop-motion . Not MIDI-mappable.
LFO Auto-Trigger	Off, On	Off	Arms a clock that takes frames on its own. Shown in Stop-motion only, and stays live during a take.
Trigger Rate	-1 to 1	0.3	How fast that clock runs, with a squared response. The default is about one capture every 1.2 seconds and full travel about nine and a half a second. Shown in Stop-motion only.
Capture Phase	0 to 1	0.5	Where in each cycle the frame is taken. Shown in Stop-motion only.

The transport is **Record**, then **Pause** and **Stop** in **Long-form**, with **Pause** becoming **Resume** once pressed. In **Stop-motion** it is **Record** and **Stop**, with a **Capture Frame** button that adds one frame and is available only while a take is running.

Screenshot writes the current frame as a PNG at **Output Resolution**. **Choose Output Folder...** picks where both go, and reads that folder's name once one is chosen.